



Topic 1

Cloud Principles

Lesson Objectives

1

Objective 1: Understand the difference between cloud computing and on-premise computing.

2

Objective 2: Learn how to access major cloud platforms via portals, APIs, and SDKs.

3

Objective 3: Explore the definition, characteristics, and advantages of cloud computing.

4

Objective 4: Understand the relationship between Cloud Computing and Service-Oriented Architecture (SOA).

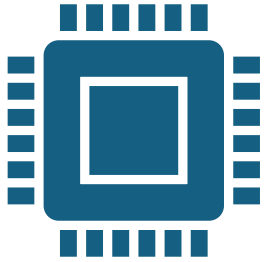
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Objective 5: Discuss enterprise cloud drivers and adoption trends.

What is Cloud Computing?

"Cloud Computing is the delivery of computing services (servers, storage, databases, networking, software, analytics, etc.) over the internet ('the cloud')."

On-Premise vs Cloud



On-Premise Computing:

Infrastructure managed internally.

Requires significant upfront capital expenditure.

Scalability is limited by hardware availability.



Cloud Computing:

Infrastructure managed externally by cloud providers.

Follows a pay-as-you-go model.

High scalability and flexibility.

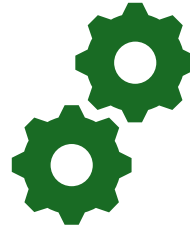
Cloud vs On-Premises

CLOUD	FEATURE	ON-PREMISES
Software is hosted and managed by a cloud provider.	Deployment	Software is installed and executed on the organization's servers.
Typically, pay-as-you-go subscription model.	Cost	Upfront capital expenditures for hardware and software, ongoing maintenance, and support costs.
It is easy to scale resources up or down as needed.	Scalability	More complicated and expensive to scale.
Applications and data can be accessed from anywhere with an internet connection.	Accessibility	Applications and data are only accessible from within the organization's network.
Cloud providers offer a wide range of security features and certifications.	Security	Security is the responsibility of the organization.
Cloud providers offer compliance solutions for a variety of industry regulations.	Compliance	The organization is responsible for ensuring compliance with all applicable regulations.
Cloud providers have a team of experts who manage and maintain the cloud infrastructure.	Expertise	The organization needs an IT team to manage and maintain its on-premises infrastructure.

Cloud Access Methods



Portals: Web-based dashboards for cloud service management (e.g., AWS Management Console).



APIs: Programmatic access for automated tasks and integrations.



SDKs: Language-specific libraries (e.g., AWS SDK for Python, Azure SDK for Java).

Cloud Computing Characteristics

Elasticity

**Multi-
Tenancy**

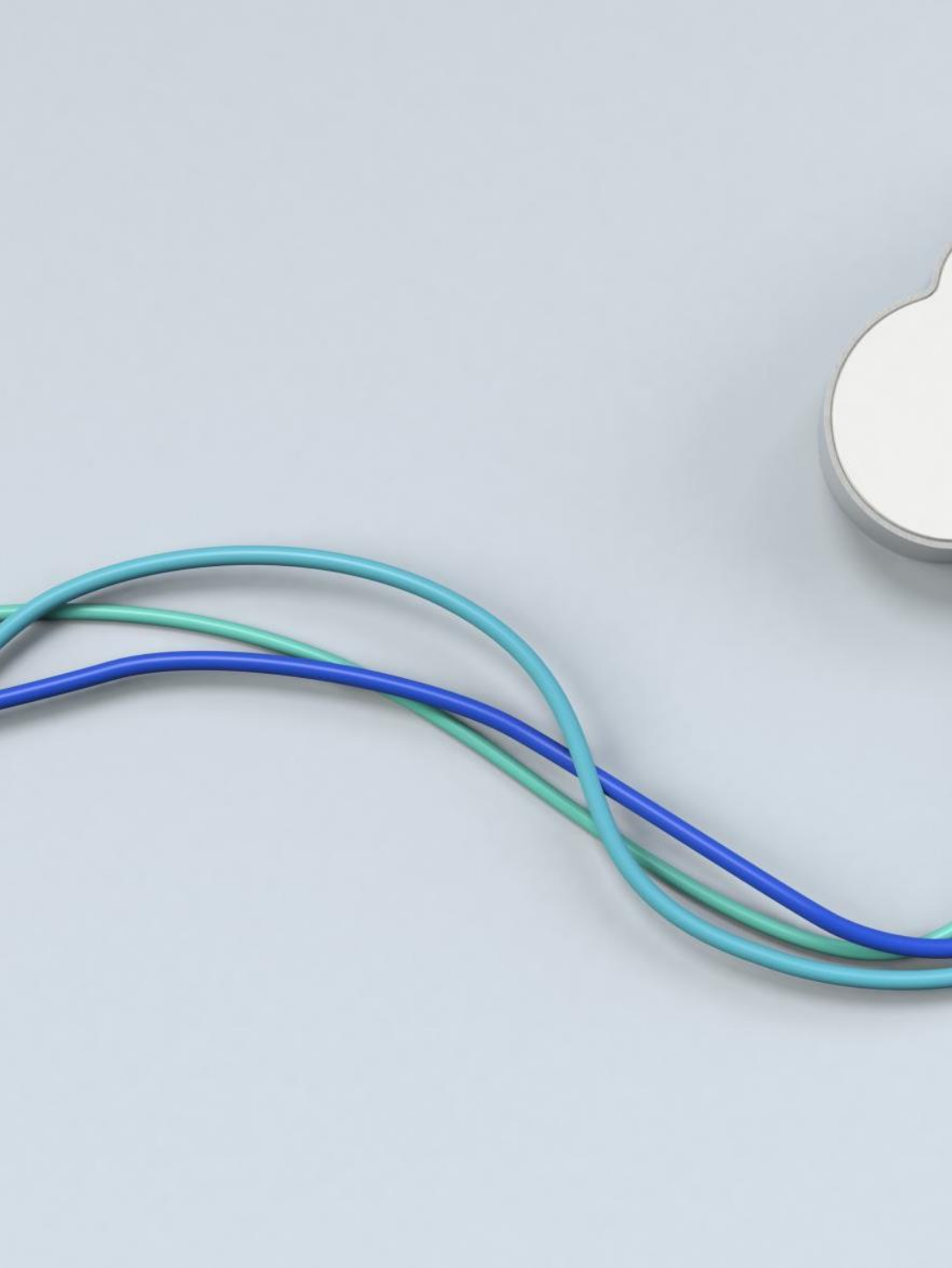
**On-Demand
Access**

**Ubiquitous
Access**

**Usage
Metering**

**Self-Service
Capability**

**SLA
Monitoring**



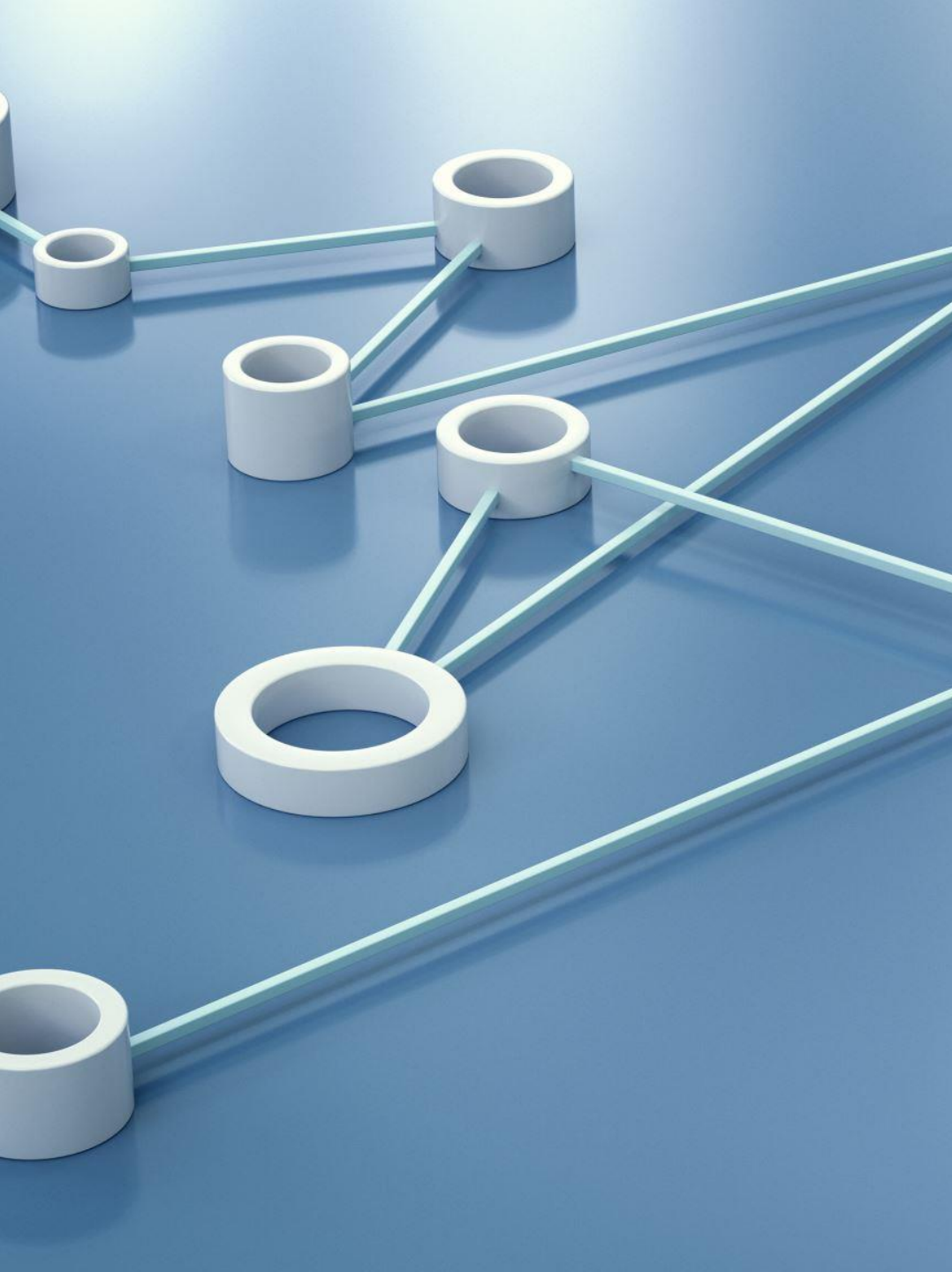
Elasticity

Elasticity refers to the ability to scale computing resources up or down dynamically based on current demand. This means if you have a sudden spike in user activity, the cloud can allocate additional resources automatically, and when demand drops, it reduces the resources to avoid wastage.

Multi-Tenancy

- Multi-tenancy is a cloud architecture where multiple users (tenants) share the same physical infrastructure while keeping their data and applications isolated and secure.
- A cloud provider like AWS hosts data for multiple organizations on the same server, but each organization's data is accessible only to them.





On-Demand Access

- Users can access and use computing resources whenever they need them, without waiting for manual provisioning.
- A developer can instantly spin up a virtual server or database instance with just a few clicks in a cloud portal or an API call.
- Immediate resource availability.
- Faster deployment of applications.
- Improved flexibility and agility for businesses.



Ubiquitous Access

- Cloud resources can be accessed from anywhere with an internet connection, using any compatible device, like a laptop, tablet, or smartphone.
- Employees working remotely can access cloud-based productivity tools like Google Workspace or Microsoft 365 from any location.

Usage Metering

- Cloud providers use metering systems to track resource usage, such as CPU, memory, storage, and bandwidth. Users are billed based on their actual consumption.
- A cloud provider charges a company for storage in GB, compute power in hours, and data transfer in TB.



Self-Service Capability

Users can independently provision and manage resources through a self-service portal without needing intervention from the service provider's support team.

A developer can create a virtual machine, configure it, and deploy an application directly from the cloud dashboard or CLI.

SLA Monitoring

Service Level Agreements (SLAs) define the expected level of service, such as uptime, performance, and response times. Cloud providers continuously monitor their systems to meet these commitments.

AWS provides an SLA of 99.99% uptime for certain services. If the uptime drops below this threshold, customers may receive compensation as per the SLA terms.

Service-Oriented Architecture (SOA)

Service-Oriented Architecture (SOA):

Framework for designing and developing software as modular services.

Relationship with Cloud Computing:

SOA principles enhance cloud computing by enabling service reuse, scalability, and flexibility.

Example: Microservices architecture commonly deployed in cloud environments.

Feature	SOA	Monolithic	Microservice	Event-Driven	Layered
Definition	Design paradigm structuring software as modular services communicating via protocols.	Entire application built as a single, unified unit.	Breaks down applications into small, highly specific services.	System communicates using asynchronous events.	Applications divided into layers like presentation, business logic, and data.
Modularity	Medium - Coarse-grained services focused on business processes.	Low - Single deployable codebase.	High - Each microservice handles a single functionality.	High - Components react to and produce events.	Medium - Organized into distinct functional layers.
Scalability	Scalable but may require more effort for inter-service communication.	Limited; scaling requires duplicating the entire application.	Highly scalable; services scale independently.	Highly scalable; can handle real-time workloads efficiently.	Limited; scalability depends on specific layer bottlenecks.
Interdependence	Loosely coupled services with some interdependencies.	Highly interdependent; components tightly coupled.	Loosely coupled with minimal interdependencies.	Minimally coupled; services interact indirectly via events.	Layers interact sequentially; moderately coupled.
Technology Flexibility	Supports heterogeneous technologies with protocols like SOAP/REST.	Requires uniform technology stack.	High flexibility; services can use different technologies.	Flexible; integrates with various messaging technologies (e.g., Kafka).	Relatively inflexible; layers depend on each other.
Use Cases	Enterprise systems, reusable services.	Small-scale applications or systems with simple requirements.	Large-scale, complex systems requiring agility (e.g., Netflix, Uber).	IoT, real-time notifications, stock trading platforms.	Traditional enterprise systems like CRM or ERP.
Advantages	Reusability, flexibility, promotes modular design.	Easy to develop initially, fewer moving parts.	Independent scalability, faster development cycles.	Low latency, resilient, real-time communication.	Clear separation of concerns, simplifies maintenance.
Challenges	Complex governance, potential performance overhead.	Hard to maintain and scale as the application grows.	Complex management; requires orchestration tools.	Debugging/tracing events is challenging; requires robust event handling.	Can introduce inefficiencies due to abstraction layers.

Enterprise Cloud Drivers



Cost reduction (CAPEX to OPEX shift).



Improved scalability and flexibility.



Faster time-to-market for applications.



Enhanced security features provided by cloud providers.



Adoption Trend

- Increasing multi-cloud strategies.
- Migration to hybrid cloud setups.
- Growing reliance on AI and ML services in the cloud.



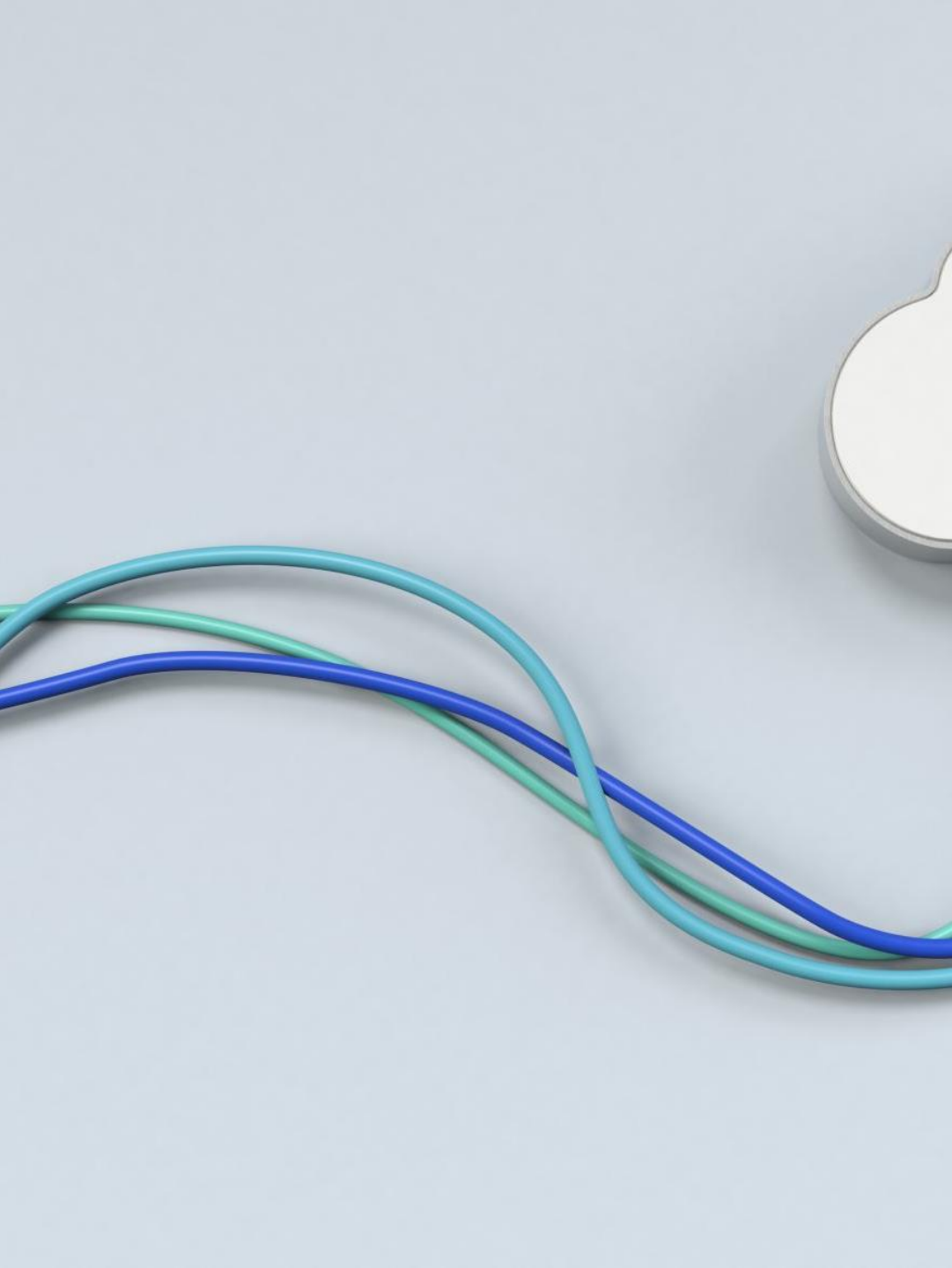
End of Topic 1





Topic 2

Architectural Layers of Cloud Computing



Contents

- **Introduction to Cloud Computing**
- **Cloud Computing Layers**
- **Software as a Service (SaaS)**
- **Platform as a Service (PaaS)**
- **Infrastructure as a Service (IaaS)**
- **Dynamic Infrastructure**
- **Summary & Q&A**

Cloud Computing?



Computers can fly now?

quickmeme.com

IF I USE CLOUD COMPUTING

**WOULD I LOSE DATA DURING
RAIN?**

Introduction to Cloud Computing

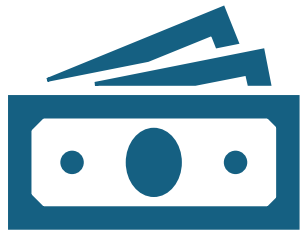
- Cloud computing is the on-demand delivery of computing services (servers, storage, databases, networking, software) over the internet.
- **Key Characteristics:**
 - On-demand self-service
 - Broad network access
 - Resource pooling
 - Rapid elasticity
 - Measured service (pay-as-you-go)



Delivery of Computing Services

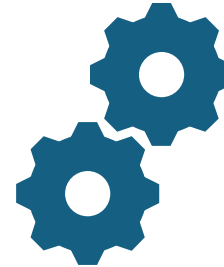
- Servers
- Storage
- Databases
- Networking
- Software
- Analytics
- Artificial Intelligence

Why Cloud Computing?



Business Advantages:

- Scalability for growing workloads
- Cost-effective model (operational vs. capital expenses)
- Faster deployment of new services
- Global reach and availability



Technical Advantages:

- Automated software updates
- Simplified infrastructure management
- Resilience and redundancy

Cloud Computing Layers

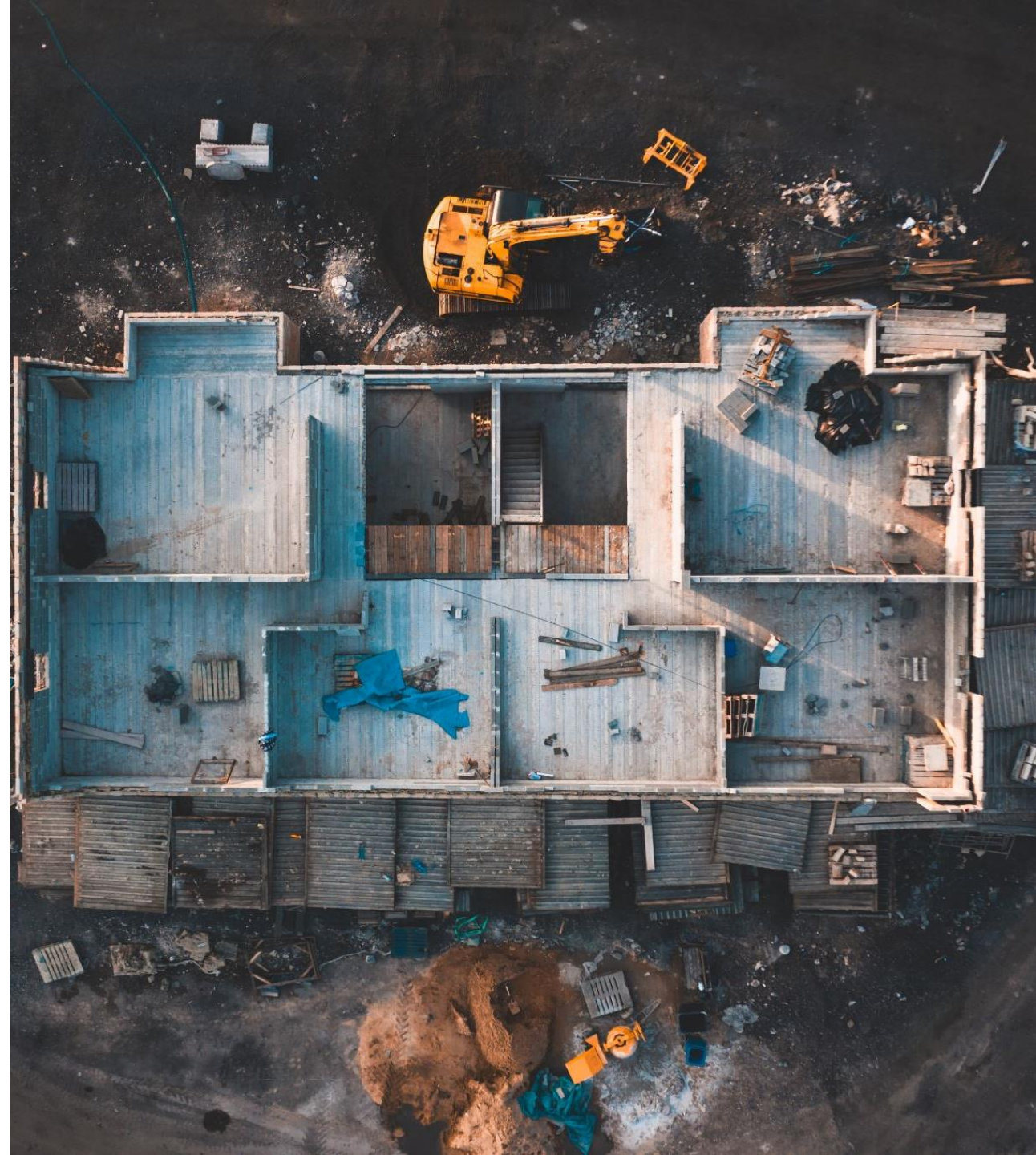
High-Level Layers in Cloud:

1. **Infrastructure as a Service (IaaS)**
 2. **Platform as a Service (PaaS)**
 3. **Software as a Service (SaaS)**
- **Goal:** Abstract complexity for end users and businesses.



Understanding the Layered Model

- **Layered Approach:**
 - Each layer builds upon the capabilities of the layer below.
 - Users at each layer only focus on resources they directly manage.
- **Analogy:**
 - SaaS: The complete package (like using a fully-furnished apartment).
 - PaaS: The foundation to build upon (like an empty apartment where you bring your own furniture).
 - IaaS: The raw materials and space (like the land and building framework).



Software as a Service (SaaS)

- Software distribution model in which applications are hosted by a third-party provider and made available over the internet.
- **Key Characteristics:**
 - No local installation required
 - Subscription or pay-per-use pricing
 - Accessible via web browser or thin client
- **Examples:**
 - Salesforce, Microsoft 365, Google Workspace (Gmail, Docs), Dropbox





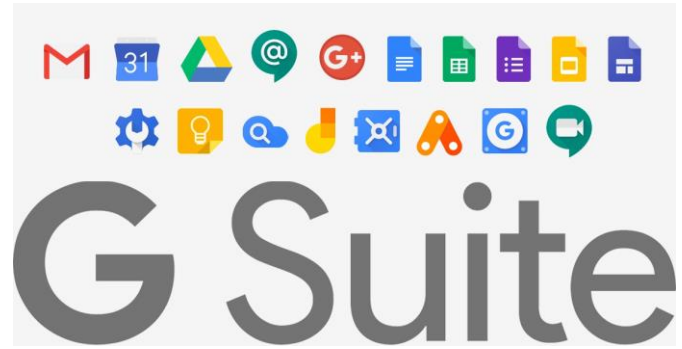
Benefits:

- Quick deployment and minimal setup
- Reduced IT overhead (maintenance, updates are handled by provider)
- Access from anywhere, on multiple devices



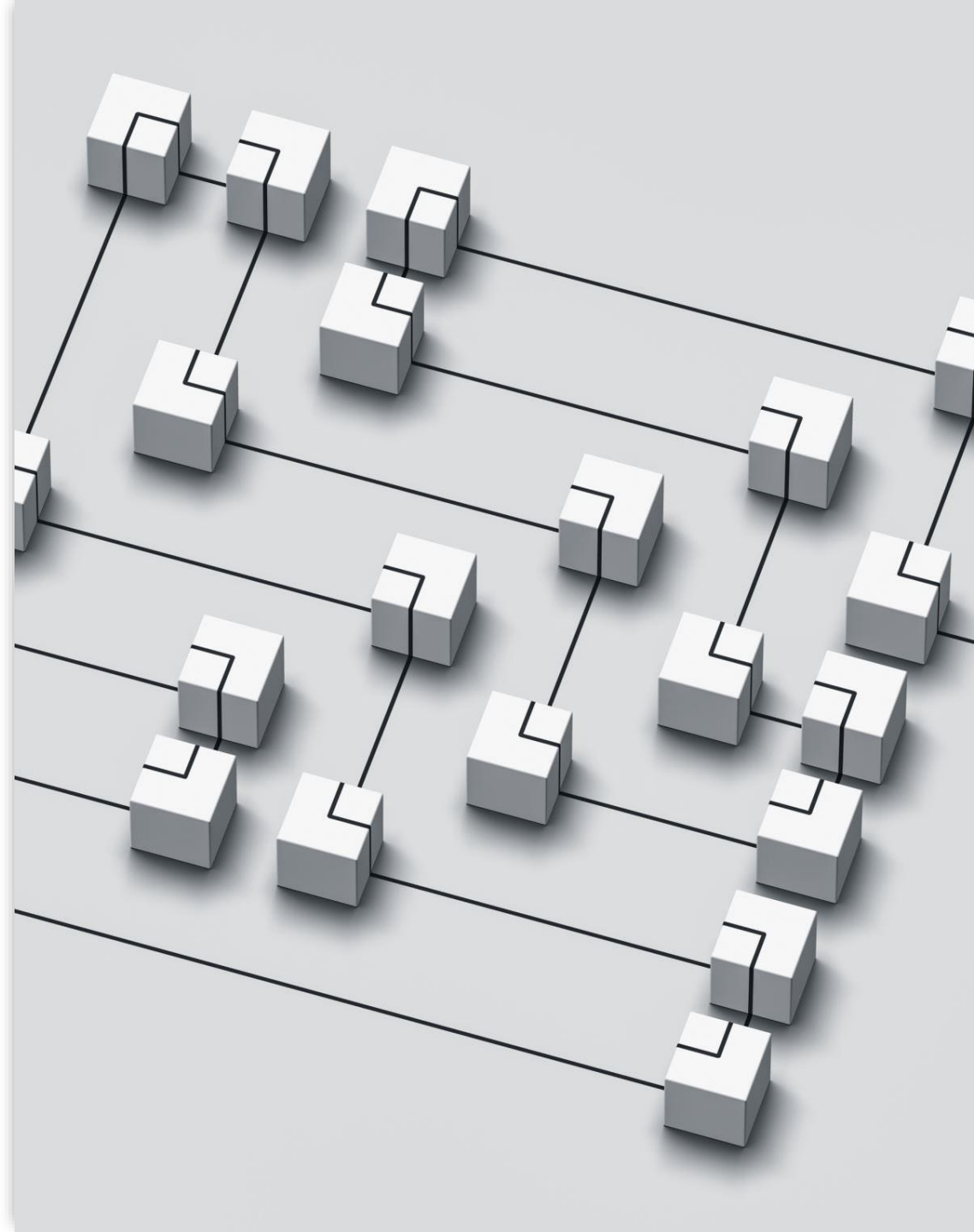
Challenges:

- Limited customization compared to on-premises software
- Dependency on internet connectivity
- Potential data privacy and compliance concerns



Platform as a Service (PaaS)

- A set of tools and services designed to make coding and deploying applications quick and efficient.
- Providers manage the underlying infrastructure; developers focus on the application itself.
- **Key Components:**
 - Runtime environment (language runtimes, frameworks)
 - Development tools
 - Databases and integration services





Benefits:

Simplifies development and deployment
Less complexity in managing servers, storage,
or networking
Automated scalability



Challenges:

Might be restricted to provider-specific
tools/languages
Risk of “vendor lock-in”
Costs can grow if the application scales
rapidly

Examples of PaaS Providers



Major PaaS Offerings:

Heroku (Salesforce)

Google App Engine (Google Cloud Platform)

Azure App Service (Microsoft Azure)

AWS Elastic Beanstalk (Amazon Web Services)



Key Services Offered:

Hosting environments

Integrated development environments (IDEs)

Database and storage integrations



Platform As A Service (PAAS)



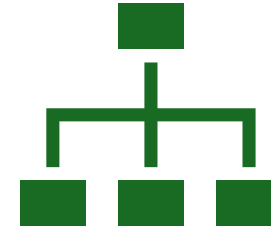
Infrastructure as a Service (IaaS)

- Virtualized computing resources over the internet, including servers, storage, and networking hardware.
- Users install and manage operating systems, applications, and middleware on provided infrastructure.
- **Key Components:**
 - Virtual machines (VMs)
 - Storage (object, block, file)
 - Networking (virtual networks, firewalls, load balancers)



Benefits:

- High flexibility and control over resources
- Pay-as-you-go model for infrastructure
- Easy scalability and rapid provisioning



Challenges:

- Requires in-house expertise to configure and manage
- Security and compliance are partly the customer's responsibility
- Potential complexity in managing virtual machines and networks

IaaS Providers and Use Cases

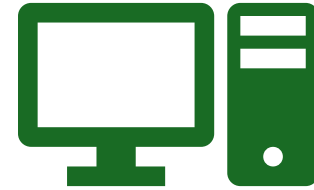


Major IaaS Providers:

Amazon Web Services (EC2, S3, etc.)

Microsoft Azure (Virtual Machines, Azure Storage)

Google Cloud Platform (Compute Engine, Cloud Storage)



Typical Use Cases:

Hosting web applications

Big data processing

Backup and disaster recovery

Development and testing environments

Infrastructure As A Service (IAAS)



Google Cloud



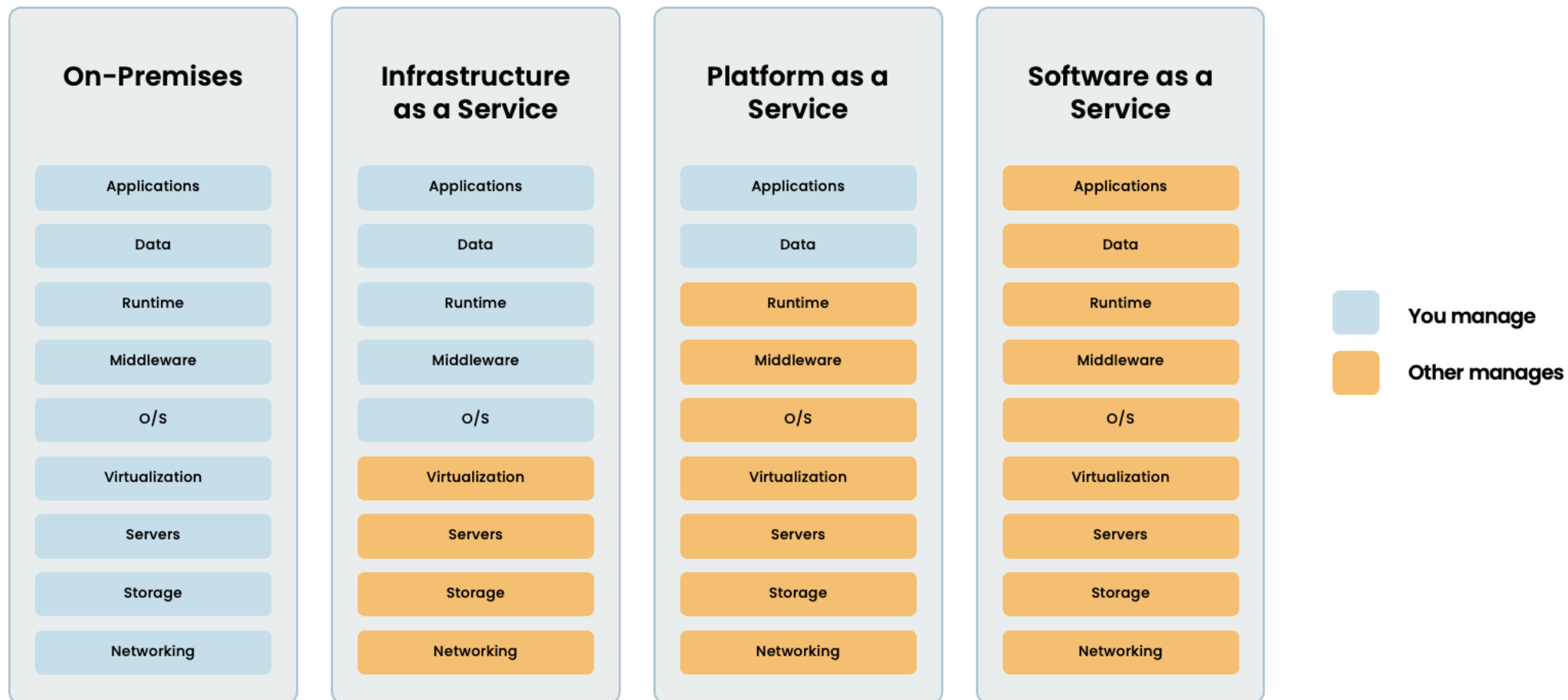
DigitalOcean

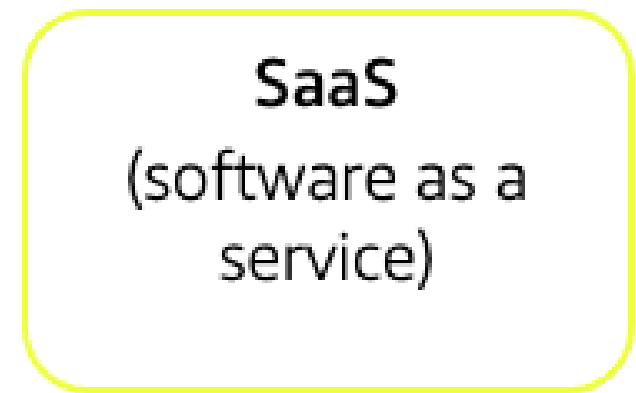
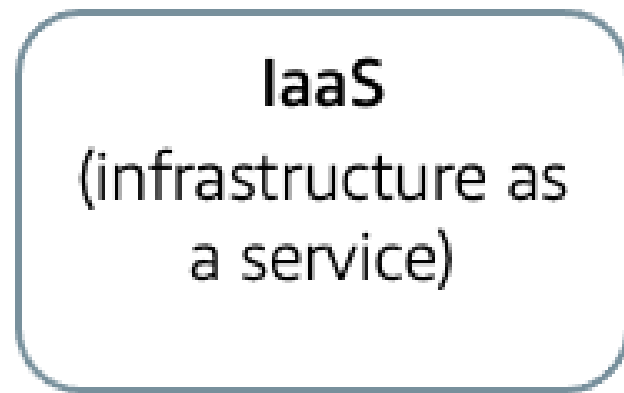


Alibaba Cloud



Microsoft Azure



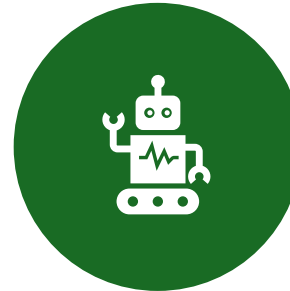


More control
over IT resources

Less control
over IT resources



The ability to quickly adapt and reconfigure IT infrastructure resources as demands change.



Often enabled by virtualization, automation, and orchestration technologies.



Ensures optimal performance, cost efficiency, and high availability



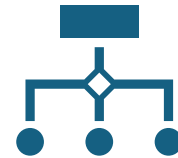
Facilitates continuous integration and continuous deployment (CI/CD)



Virtualization - Abstracts hardware into virtual resources (VMs or containers)



Automation - Scripts or tools to streamline provisioning, configuration, and scaling



Orchestration - Systems (like Kubernetes, Docker Swarm) that manage the lifecycle of multiple resources



Monitoring & Management - Tools to observe and adjust resource usage in real time (e.g., AWS CloudWatch, Azure Monitor)

Benefits of a Dynamic Infrastructure

Scalability

- Easily add or remove resources in response to traffic or workload changes

Agility

- Faster deployment of new applications and updates

Cost Optimization

- Pay only for the resources actually in use

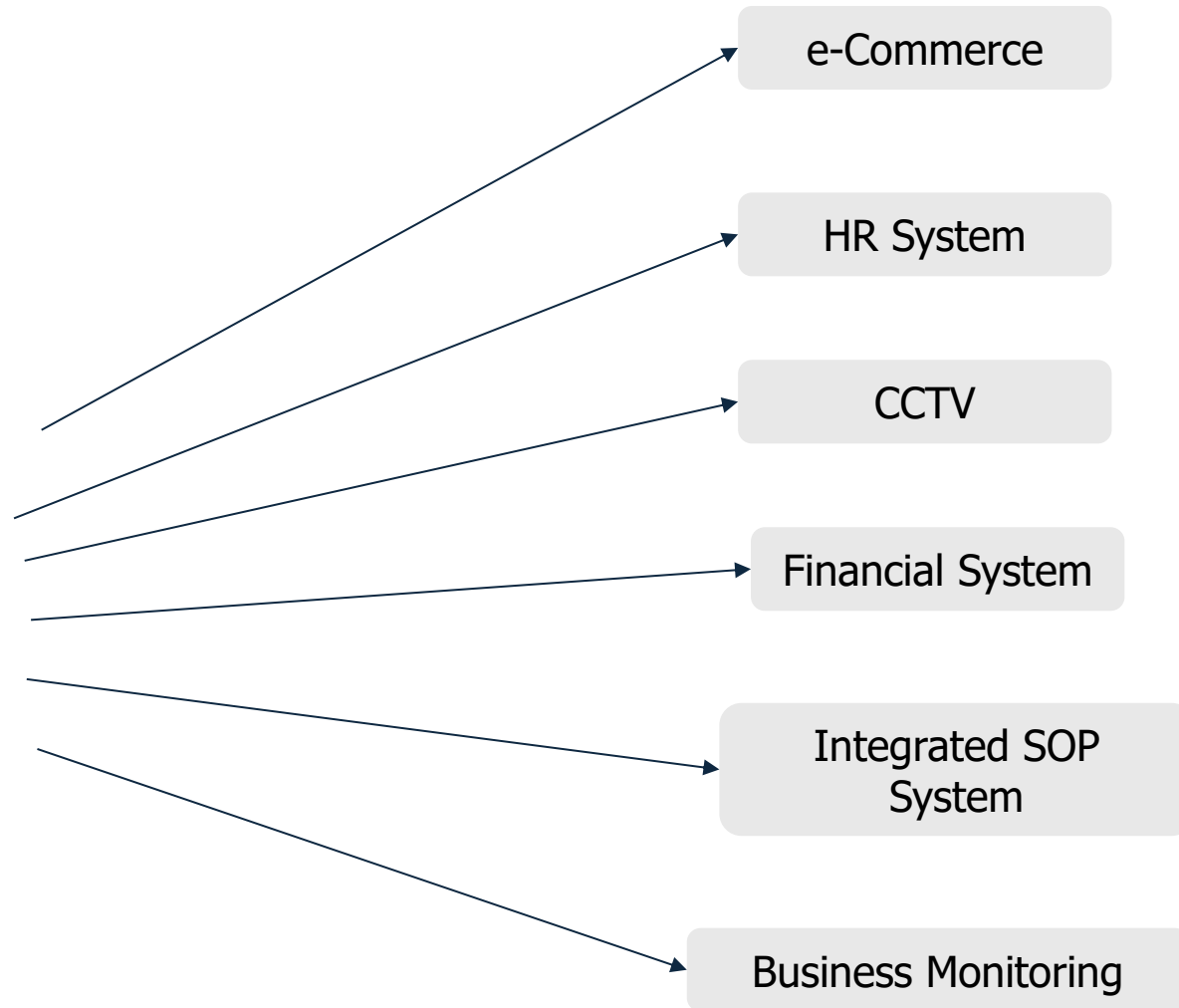
Fault Tolerance

- Automatically shift workloads if a server or region fails

Izrulfizal Shoe Store



HQ & Izrulfizal
Shoe Store



Izrulfizal Shoe Store



HQ & Izrulfizal
Shoe Store



Traditional way:

- Weeks/months to setup the infrastructure
- Running business on paper

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Shoe Store



Modern way:

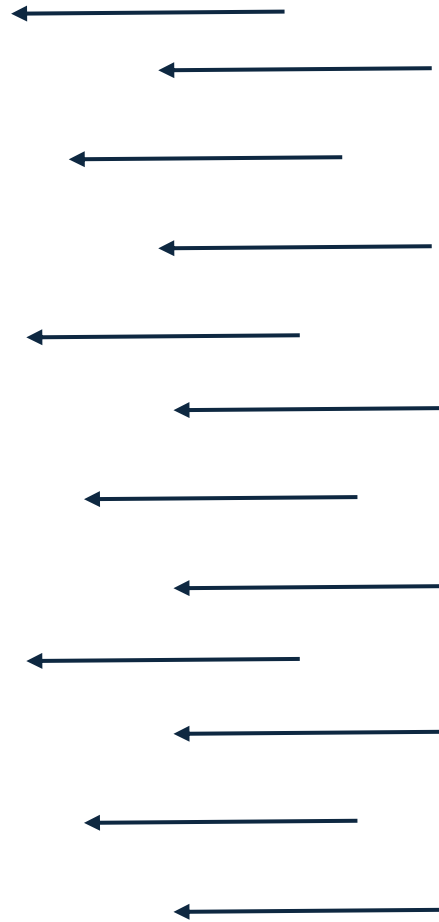
- Cloud hosting
- Server deployment in minutes
- Everything integrated
- Easy accessibility anytime

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Shoe Store

Heavy Traffics



Scalable

Vertical Scaling

(Increase size of instance (RAM , CPU etc.))



Horizontal Scaling

(Add more instances)





Complexity

Multiple layers of abstraction can be difficult to manage



Skill Gap

Advanced knowledge required for automation/orchestration tools



Governance

Ensuring compliance and security policies in rapidly changing environments



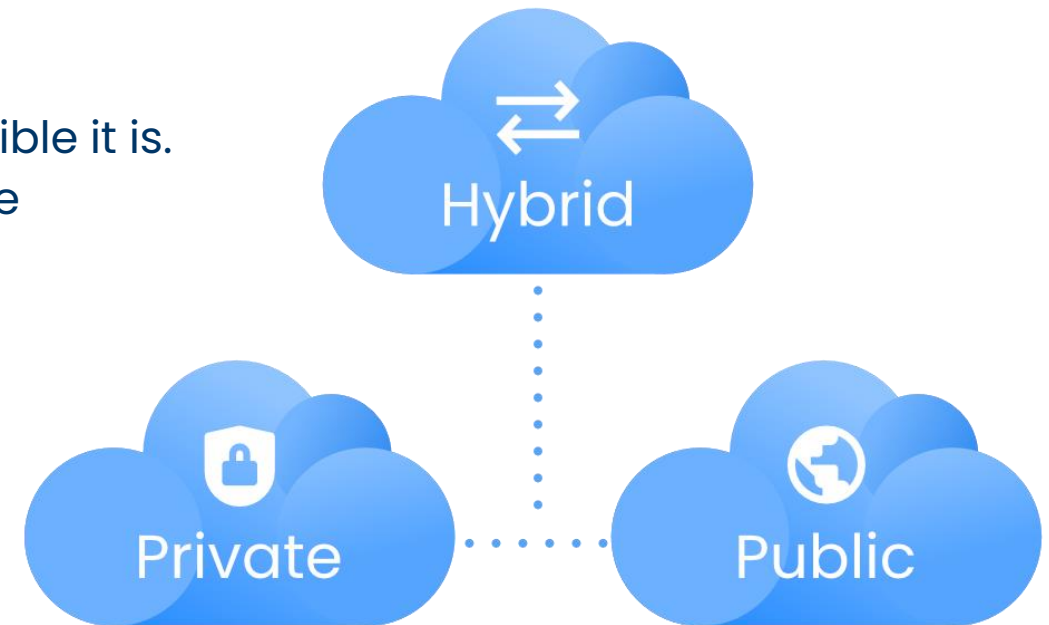
Monitoring & Debugging

Dynamic scaling can make troubleshooting more complex

Type of Cloud Deployment

Another thing about Cloud Computing is how accessible it is. For this, we can differentiate cloud computing in three different ways :

- **Public Cloud**
- **Private Cloud**
- **Hybrid Cloud –**
 - **Mixture of Private & Public**



Putting It All Together

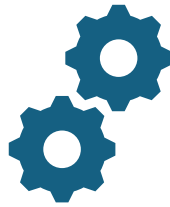


Layered Approach Recap

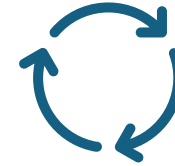
SaaS: Complete software solutions provided to end users

PaaS: Platform that abstracts infrastructure and offers tools for development

IaaS: Core infrastructure (servers, storage, networking) delivered virtually



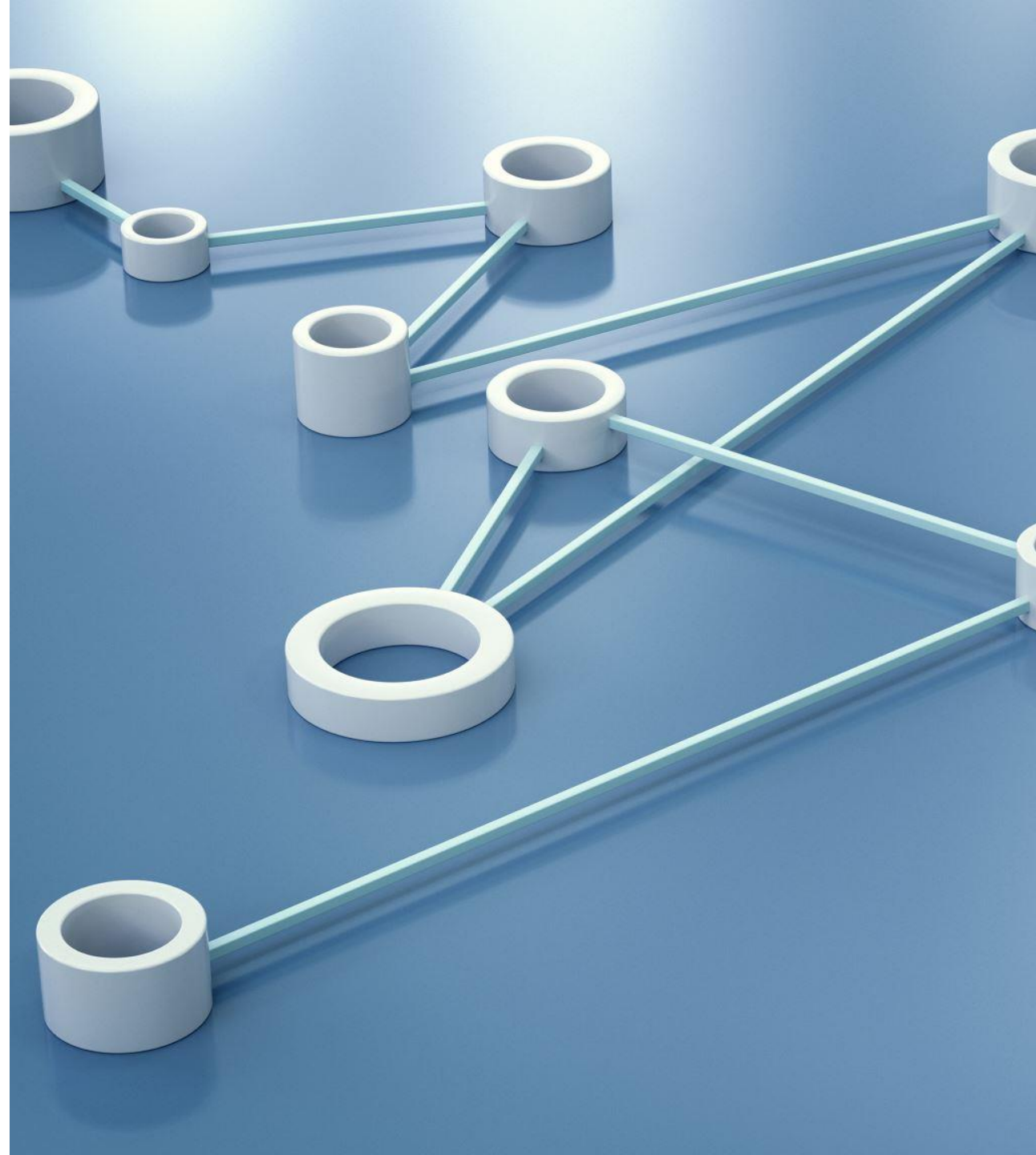
Dynamic Infrastructure - Underpins all these layers with flexibility and automation

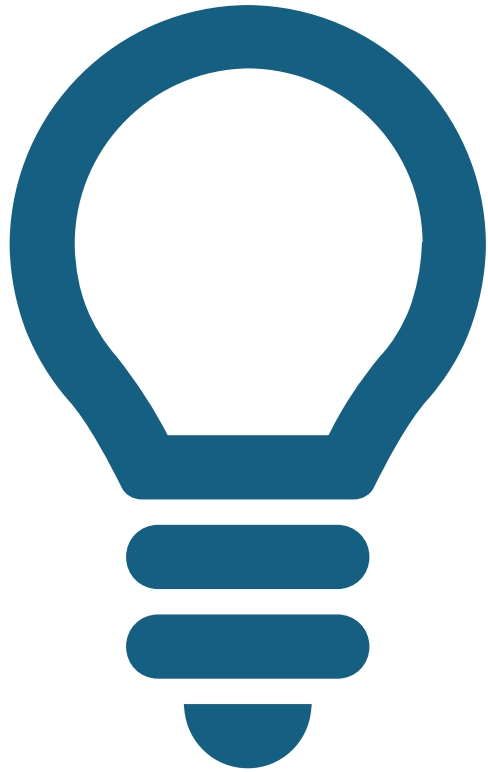


Choosing the Right Model - Depends on your business needs, development expertise, and desired level of control

Summary

- Cloud computing offers different service models (SaaS, PaaS, IaaS) to suit various needs.
- Dynamic infrastructure enables scalability, agility, and cost-effectiveness.
- Understand the trade-offs (control vs. complexity, cost vs. flexibility).





**System Models and Enabling
Technologies**

Topic 3

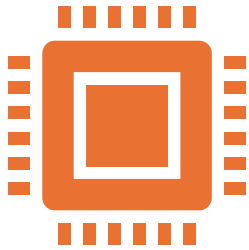
Lecture Outline

- System Scalability
- Amdahl's Law
- System Availability
- Single Point of Failure
- Security Threats and Defense Technologies
- Types of Attacks and Potential Damages
- System Attacks and Network Threats
- Internet Security Responsibilities
- Case Studies

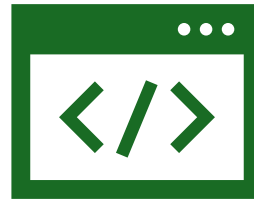
System Scalability

- The capability of a system to handle an increasing load or demand without performance degradation.
- Importance
 - Critical for system growth
 - Improved performance
 - Long-term sustainability.





Vertical Scalability: Enhancing existing resources, such as adding more CPU, RAM, or storage.



Horizontal Scalability: Expanding by adding more servers or devices.



Elastic Scalability: Automatically adjusting resources in response to demand.

Benefits of Scalability



Ensures consistent performance during peak loads.



Reduces system downtime and service disruptions.



Optimizes operational costs by scaling resources as needed.



Supports business growth and user demand.

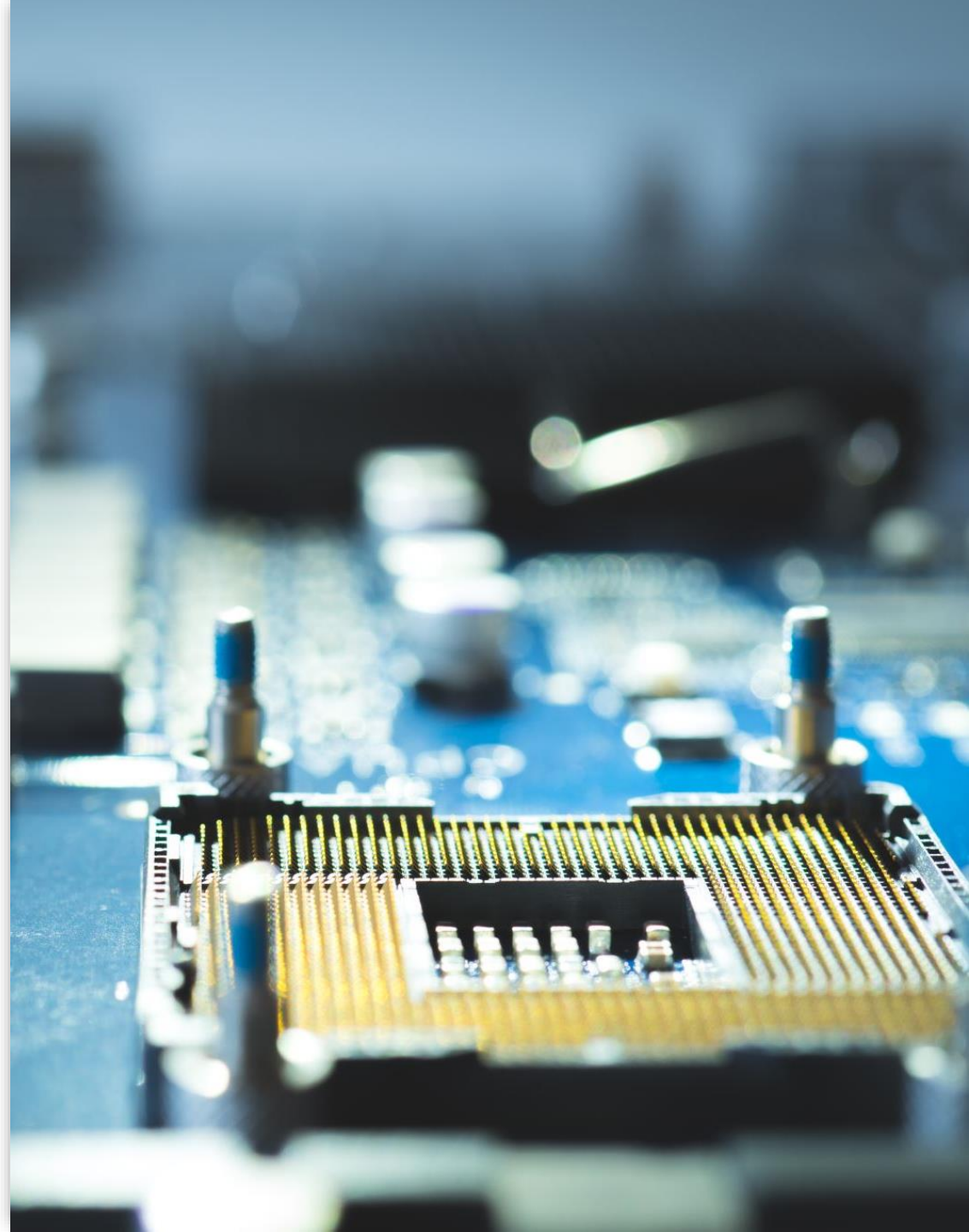
Amdahl's Law

- Describes the limitations of speedup in parallel computing due to the serial portion of a task.
- Formula: $\text{Speedup} = 1 / (S + (1 - S)/N)$
 - S = Fraction of the workload that must be executed serially
 - N = Number of processors



Amdahl's Law

- Scenario: A task where 40% is serial and 60% can be parallelized across 4 processors.
 - $\text{Speedup} = 1 / (0.4 + (0.6/4)) = 2.22x$ faster.
- Increasing processors has diminishing returns if the serial portion is significant.

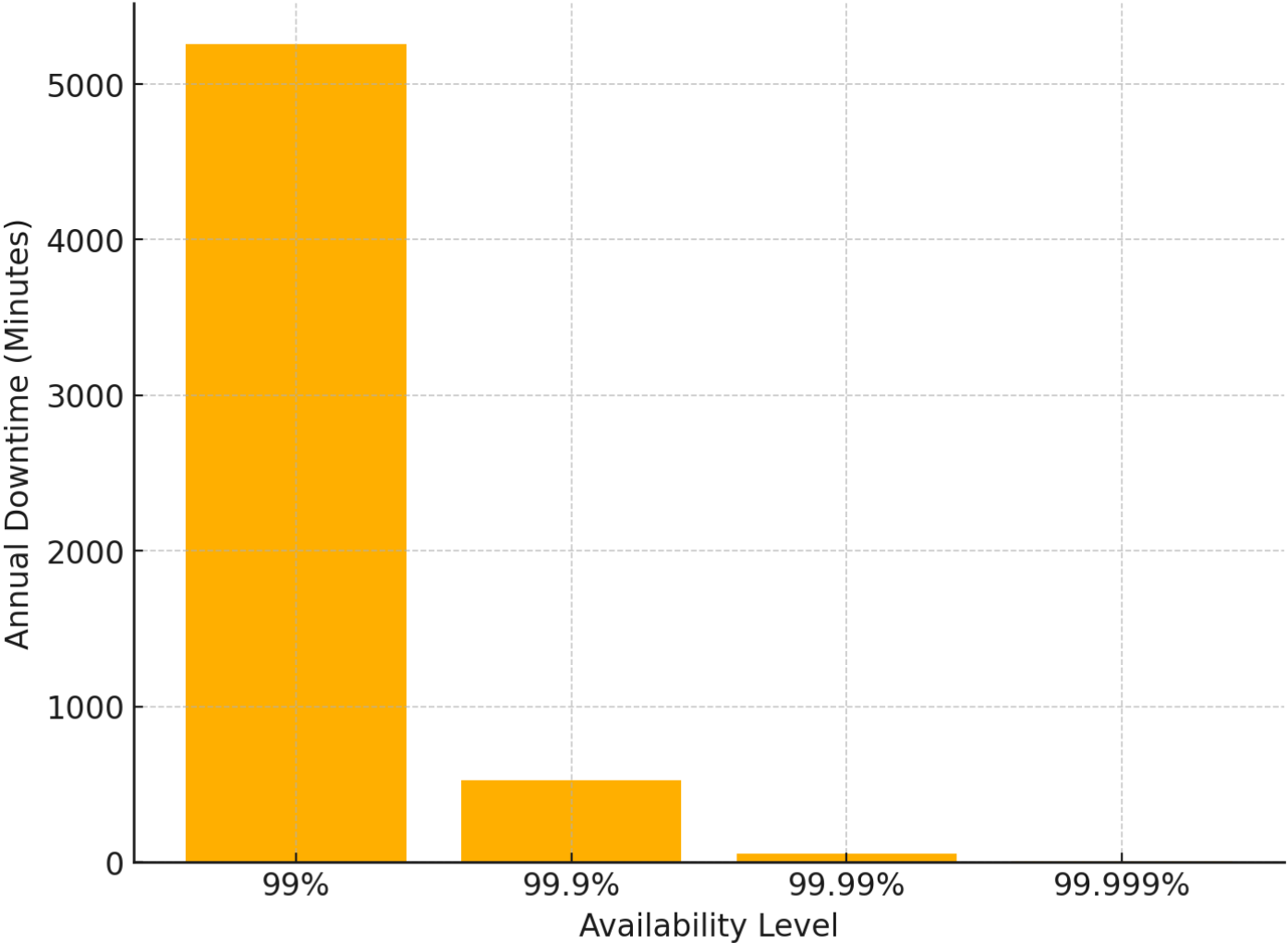


System Availability

- The degree to which a system is accessible and functional when needed.
- Measured as: $\text{Availability (\%)} = (\text{Uptime} / \text{Total Time}) * 100$



Annual Downtime for Different Availability Levels





Hardware failures



Software bugs and
crashes



Network
disruptions



Maintenance and
upgrades



Disaster events
(natural disasters,
cyberattacks)

Ensuring High Availability



Redundancy: Use backup components and systems.



Load Balancing: Distribute traffic to avoid overloading servers.



Failover Mechanisms: Automatically switch to standby systems during failures.



Monitoring: Continuous monitoring for quick issue detection.

A component whose failure
leads to total system
shutdown.



Examples:

Single
database
server

Central router
in a network

Power supply
without backup



Use redundant components and backup systems.



Implement clustering and failover solutions.



Distribute services across multiple data centers.

Threats: Actions or events that compromise system integrity, confidentiality, or availability.

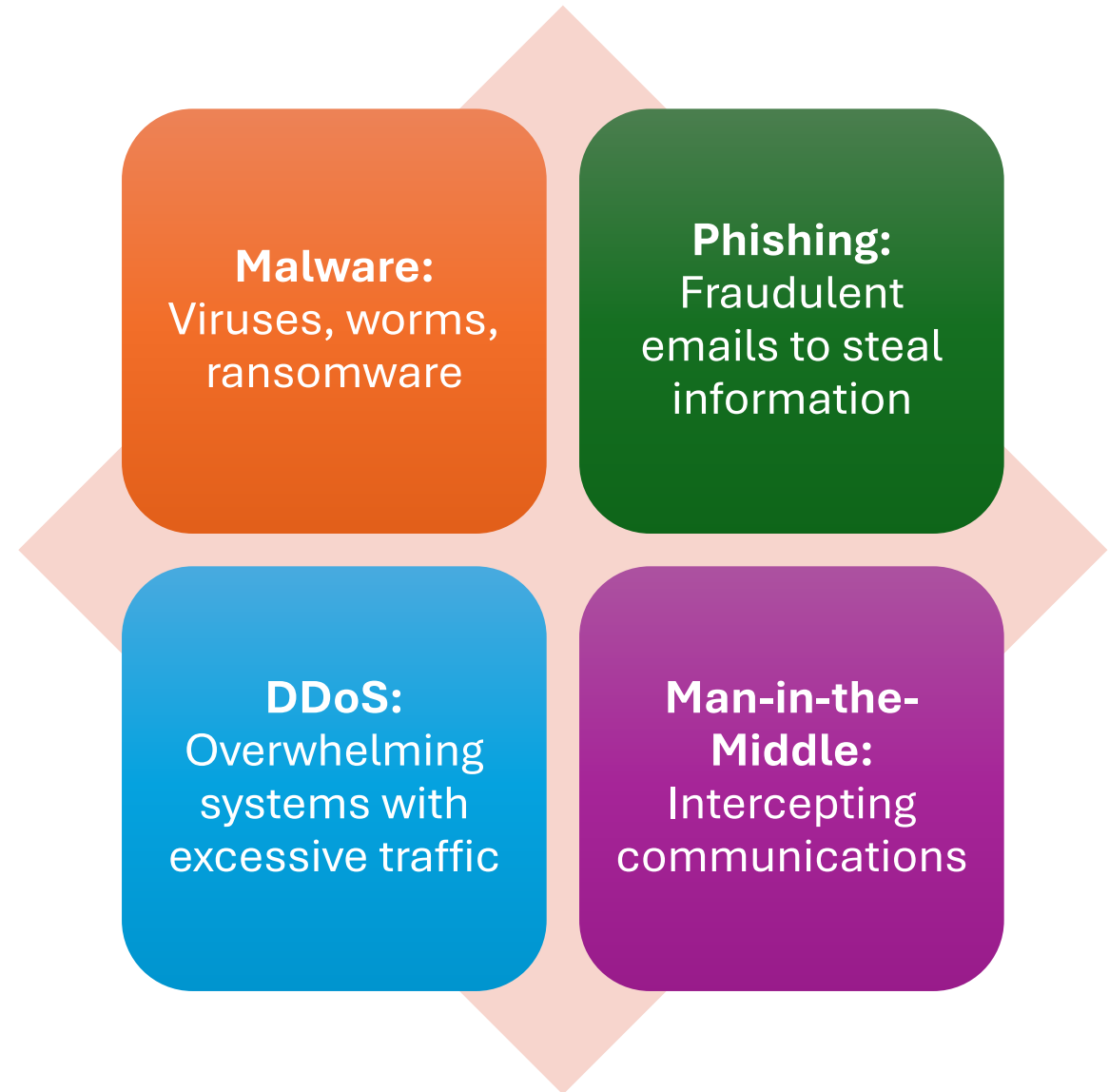


Defense Technologies:

Firewalls filter network traffic.

Intrusion Detection Systems (IDS) monitor suspicious activity.

Encryption secures data during transmission and storage.





Data breaches leading
to loss of sensitive
information



Financial losses due to
business disruption



Reputational damage
affecting customer trust



Legal consequences
from non-compliance

System Attacks



Unauthorized Access: Gaining entry without permission



Privilege Escalation: Increasing access to sensitive areas



Insider Threats: Employees misusing access



Application Vulnerabilities: Exploiting bugs in software

Network Threats to Cyberspace



PACKET SNIFFING:
CAPTURING DATA PACKETS
DURING TRANSMISSION



IP SPOOFING:
IMPERSONATING TRUSTED
DEVICES



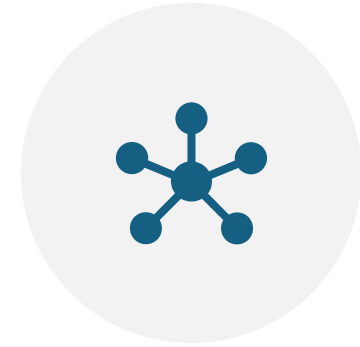
DNS POISONING:
REDIRECTING TRAFFIC TO
MALICIOUS SITES



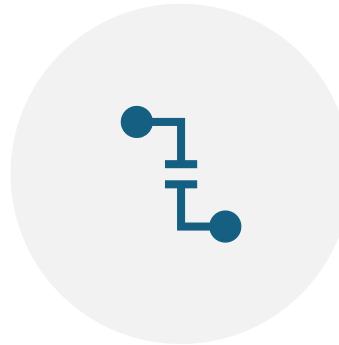
ZERO-DAY EXPLOITS: ATTACKS
EXPLOITING UNDISCLOSED
VULNERABILITIES



SECURE PROTOCOLS: USE
HTTPS AND VPNS FOR
ENCRYPTION



**NETWORK
SEGMENTATION:** ISOLATE
SENSITIVE SYSTEMS



FIREWALLS AND IPS:
BLOCK UNAUTHORIZED
TRAFFIC



PATCH MANAGEMENT:
REGULAR SOFTWARE
UPDATES

Internet Security Responsibilities

Shared responsibility between
individuals, organizations, and
governments



Responsibilities of Individuals

- Use strong, unique passwords
- Enable two-factor authentication (2FA)
- Avoid clicking on suspicious links

Responsibilities of Organizations

- Develop comprehensive cybersecurity policies
- Conduct regular employee training
- Use encryption to protect data



Responsibilities of Governments

- Enforce cybersecurity regulations and compliance
- Establish national cybersecurity frameworks
- Collaborate with international cybersecurity agencies



Case Study 1 - Equifax Data Breach

- **Event:** In 2017, hackers stole sensitive information of 147 million individuals.
- **Cause:** Vulnerability in a web application framework
- **Impact:** Massive financial loss, legal actions, and reputational damage
- **Lesson:** Regular vulnerability assessments and timely patching are critical.



Case Study 2 - WannaCry Ransomware Attack

- **Event:** In 2017, a ransomware worm infected systems globally, encrypting data and demanding ransom payments.
- **Cause:** Exploited a vulnerability in outdated Windows systems
- **Impact:** Affected hospitals, businesses, and government agencies worldwide
- **Lesson:** Keep systems updated and use robust backup solutions



Summary

- Scalability and availability are essential for system performance
- Amdahl's Law limits parallel computing benefits
- Security threats must be countered with advanced defense technologies
- Case studies highlight the importance of proactive cybersecurity

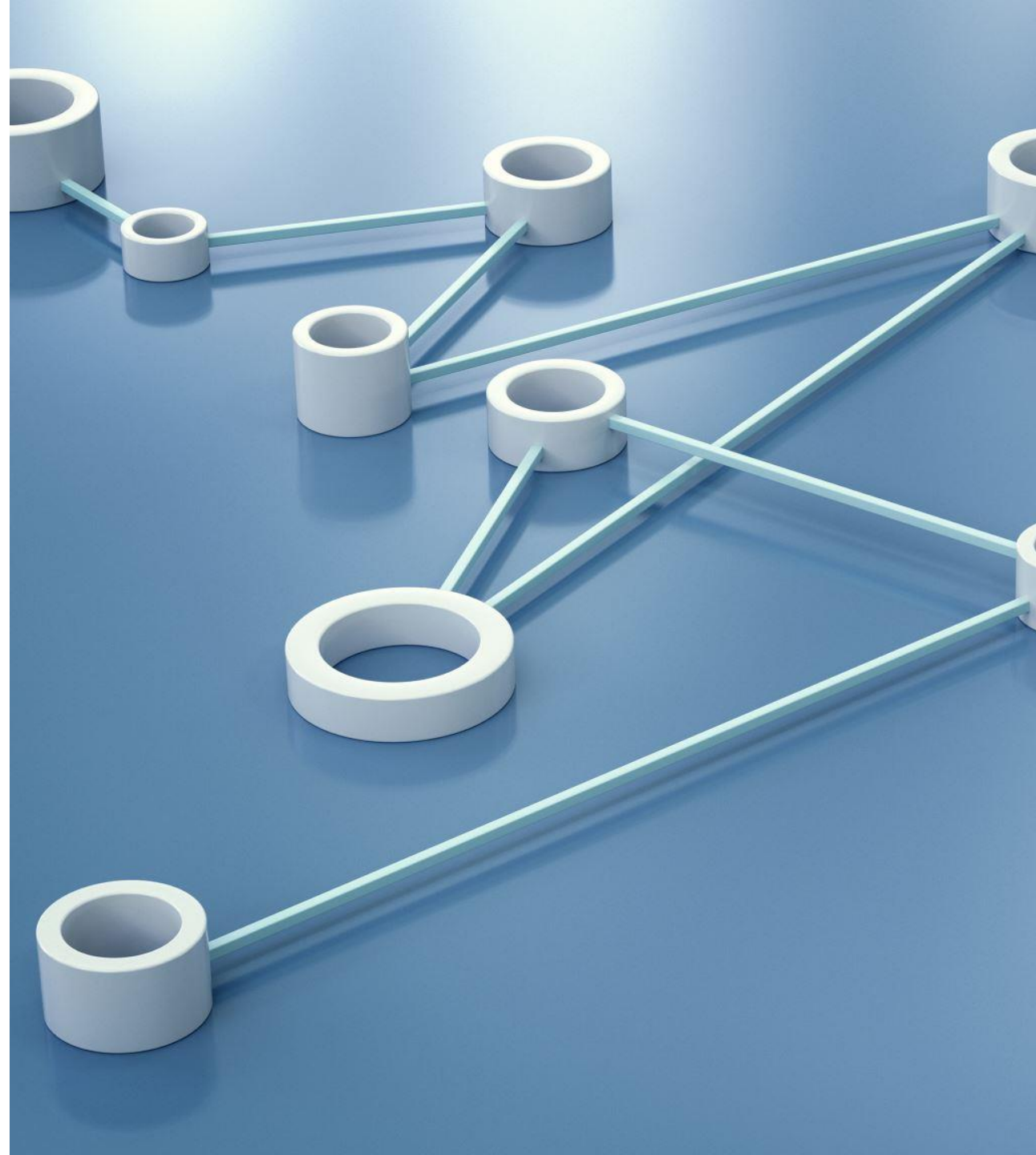


Understanding the SaaS Model,
Examples, and Areas of Application

Topic 4 – Software as a Service (SaaS)

Introduction to SaaS

- SaaS (Software as a Service) is a cloud computing model where applications are hosted by a provider and made available to users over the internet.
- Key features:
 - No need for installation
 - Subscription-based pricing
 - Access from any device with internet
 - Provider manages infrastructure and updates



Evolution of Software Delivery



Traditional Model: Buy a license, install software on a local machine.



SaaS Model: Access software through a web browser, pay as you go.

SaaS Architecture



Multi-Tenancy: A single instance serves multiple customers.



Scalability: Can handle thousands of users without performance drop.



Automatic Updates: No manual upgrades needed.



Security & Compliance: Data encryption and access control managed by the provider.

SaaS vs Other Cloud Models

Feature	SaaS	PaaS	IaaS
Example	Gmail, Dropbox	Heroku, Google App Engine	AWS EC2, DigitalOcean
User Control	Least	Medium	Most
Maintenance	Managed by provider	User manages applications	User manages OS & apps
Use Case	Software applications	App development	Infrastructure for hosting

Benefits of SaaS

Cost-Effective: No upfront software purchase.

Accessibility: Use from any device, anywhere.

Scalability: Grow or shrink user base easily.

Security: Handled by professionals.

No Maintenance: No need to update or patch software.



INTERNET DEPENDENCY:
NO INTERNET = NO ACCESS.



LIMITED CUSTOMIZATION:
SOME SAAS APPS HAVE
RESTRICTIONS.



SECURITY CONCERNS:
DATA IS STORED BY A THIRD-
PARTY PROVIDER.



VENDOR LOCK-IN:
DIFFICULT TO MIGRATE TO
ANOTHER PROVIDER.



**SUBSCRIPTION-BASED
(MONTHLY/YEARLY) – NETFLIX,
OFFICE 365**



**PAY-AS-YOU-GO (USAGE-
BASED) – AWS LAMBDA, TWILIO**



**FREEMIUM (BASIC FREE, PAID
PREMIUM FEATURES) –
DROPBOX, SLACK**



Productivity SaaS

Google Workspace (Docs, Sheets, Slides)

Microsoft 365

Dropbox



CRM SaaS

Salesforce

HubSpot

Zoho CRM



Communication SaaS

Slack

Zoom

Microsoft Teams



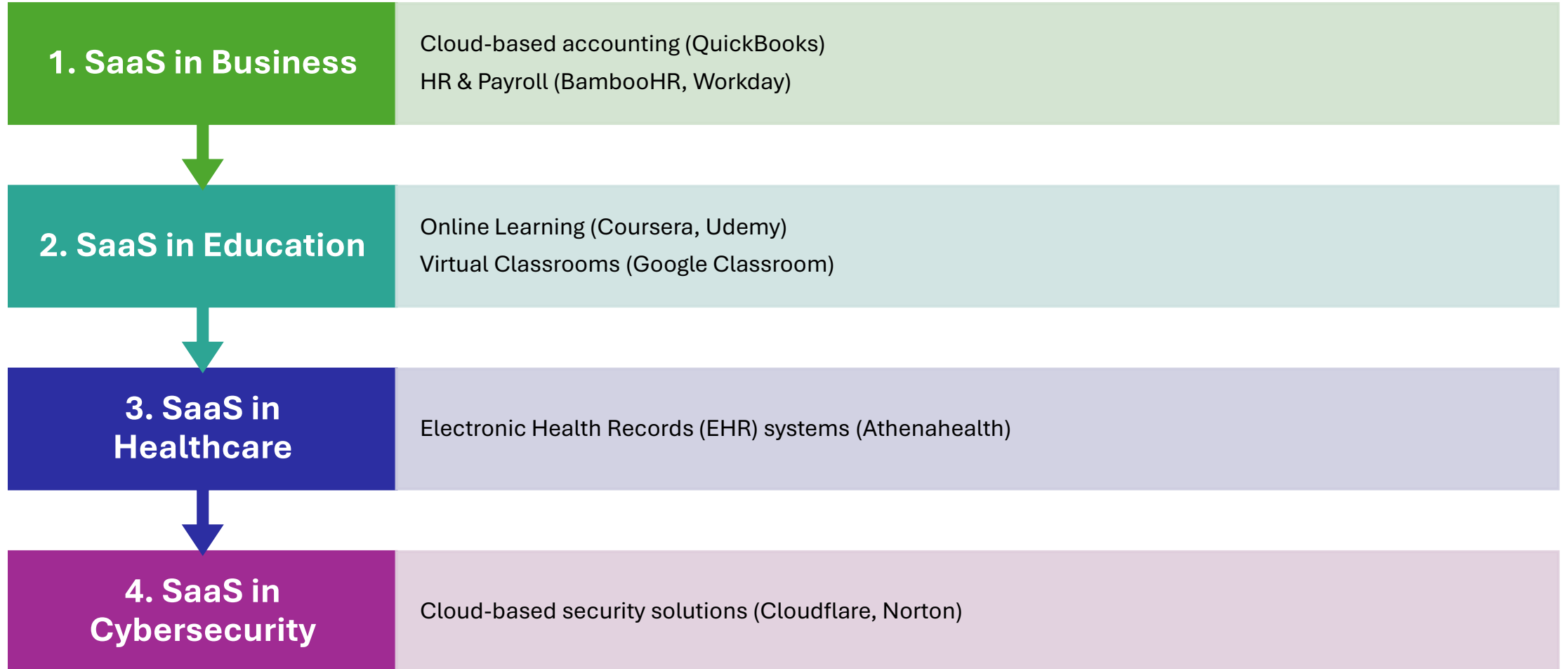
E-commerce SaaS

Shopify

BigCommerce

Wix

SaaS in Different Areas

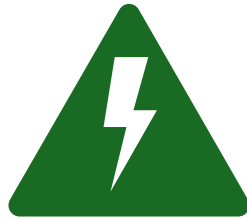


- **Factors to Consider:**
 - Pricing & Scalability
 - Security & Compliance
 - Integrations with other software

Risks & Security Concerns



Data Privacy: Ensure GDPR & compliance.



Downtime Risks: Choose reliable providers.



Cybersecurity Threats: Use MFA, encryption.

End of Topic 4

"The cloud is about how you do computing, not where you do computing."

— Paul Maritz, Former CEO of VMware

The background of the slide features a stylized, low-poly illustration of clouds. The clouds are composed of various shades of blue and white, giving them a three-dimensional, geometric appearance. They are scattered across a solid light blue background. On the left side, there is a white, torn-paper-like edge that separates the text area from the cloud illustration.

Topic 5 – Cloud Infrastructure Management

Managing the Invisible Backbone of the Cloud

Introduction to Cloud Infrastructure

Cloud infrastructure includes:

- Compute (VMs, containers)
- Storage (block, object, file)
- Network (virtual switches, routers)
- Managed via web portals or APIs

Think of it like renting a fully furnished apartment. You don't own the building (hardware), but you get everything you need inside (VM, storage, etc.)



Why Cloud Infrastructure Management Matters

- Ensures resources are:
 - Available
 - Secure
 - Efficient
- Prevents downtime and optimizes cost
- **Real-life example:**
Netflix uses cloud infrastructure to stream to millions. Without solid management, you'd be watching a spinning buffer wheel all day.



What is a Virtual Infrastructure Manager (VIM)?

- Software that manages virtualized resources in a cloud

Examples: OpenStack, VMware vSphere, Proxmox, Eucalyptus

Acts as the “invisible caretaker” of virtual resources



Provisioning
VMs



Networking
configuration



Storage
management



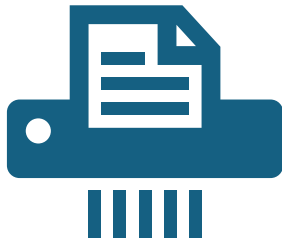
Resource
scheduling



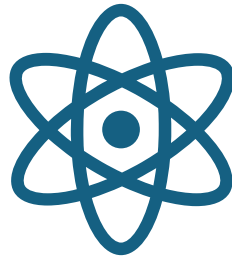
Monitoring
and logging



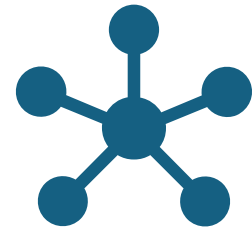
APIs for
automation



Open-source VIM



Modular: Nova (compute),
Neutron (network), Cinder
(storage)



Scalable and customizable

VMware vSphere as a VIM



ENTERPRISE-GRADE



INTUITIVE GUI



INTEGRATION WITH OTHER
VMWARE TOOLS (VCENTER,
VSAN)

Feature Comparison of VIMs

Feature	OpenStack	VMware vSphere	Proxmox
Open-source			
GUI	Moderate	Excellent	Good
Enterprise support	Moderate	Excellent	Moderate
Scalability	High	High	Medium

Advantages of Using Virtual Machines



ISOLATION
BETWEEN
WORKLOADS



BETTER
UTILIZATION OF
HARDWARE



SNAPSHOT &
ROLLBACK



DISASTER
RECOVERY



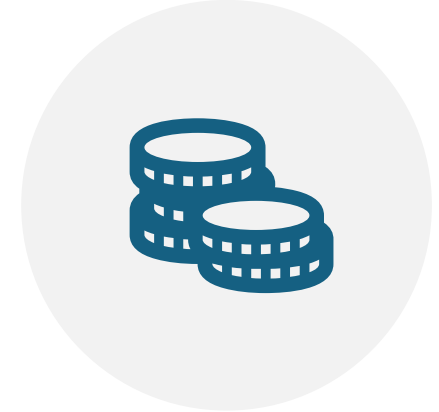
PORTABILITY



RUN MULTIPLE OS ON
ONE PHYSICAL MACHINE



SIMULATE DIFFERENT
ENVIRONMENTS



COST-EFFICIENT
DEPLOYMENT

Benefits of Virtual Machines in Cloud



Scalability: Clone in minutes



Load balancing: Migrate VMs based on demand



Maintenance: Update without downtime

Opportunities in Cloud Infrastructure Management



Automated scaling



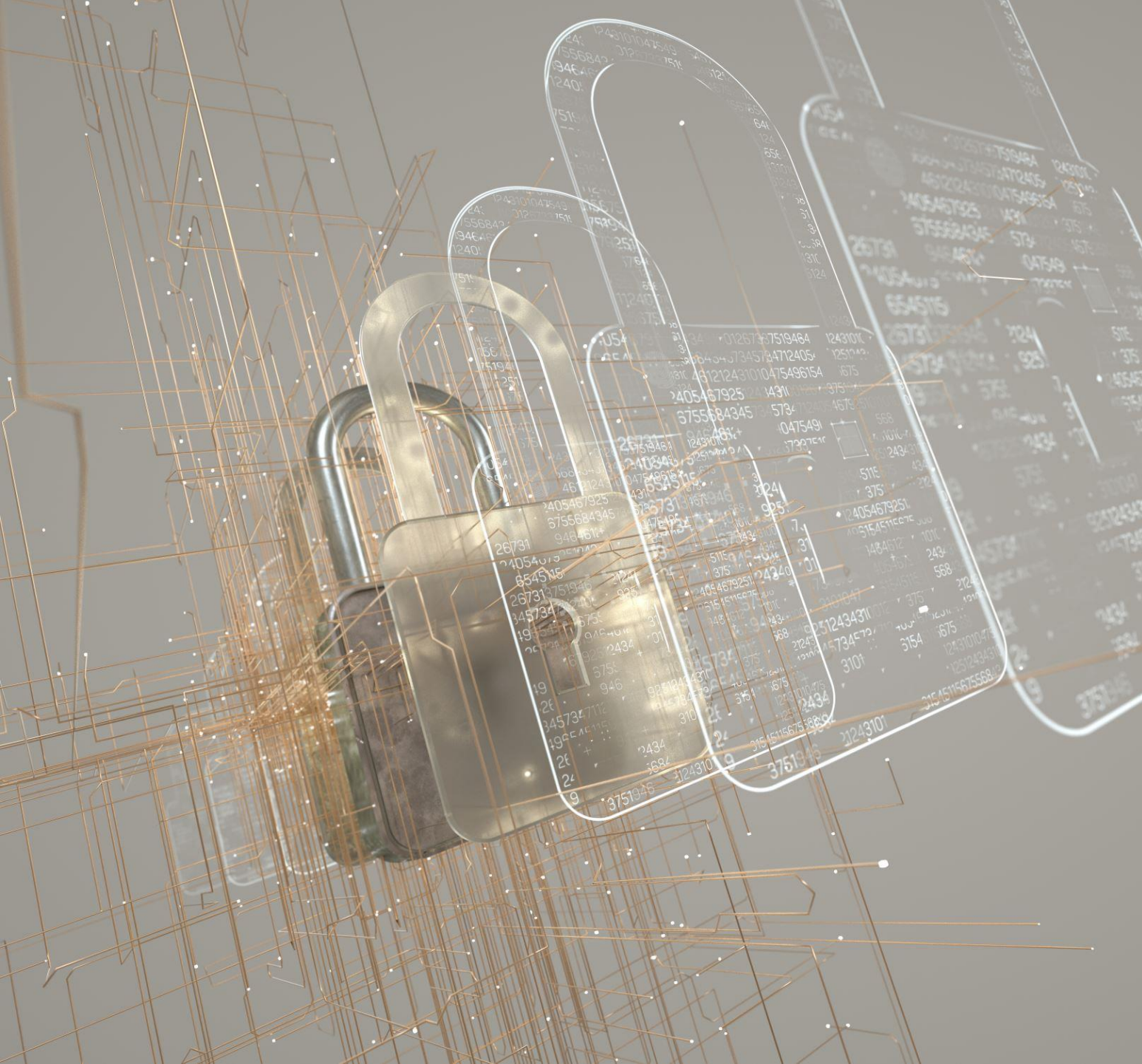
Multi-cloud
strategy



AI-based resource
optimization



Integration with
DevOps (CI/CD)



Challenges in Cloud Infrastructure

- Vendor lock-in
- Complex configurations
- Latency issues
- Skills gap

Security Issues in Virtual Infrastructure



Hypervisor attacks



VM escape



Insecure APIs

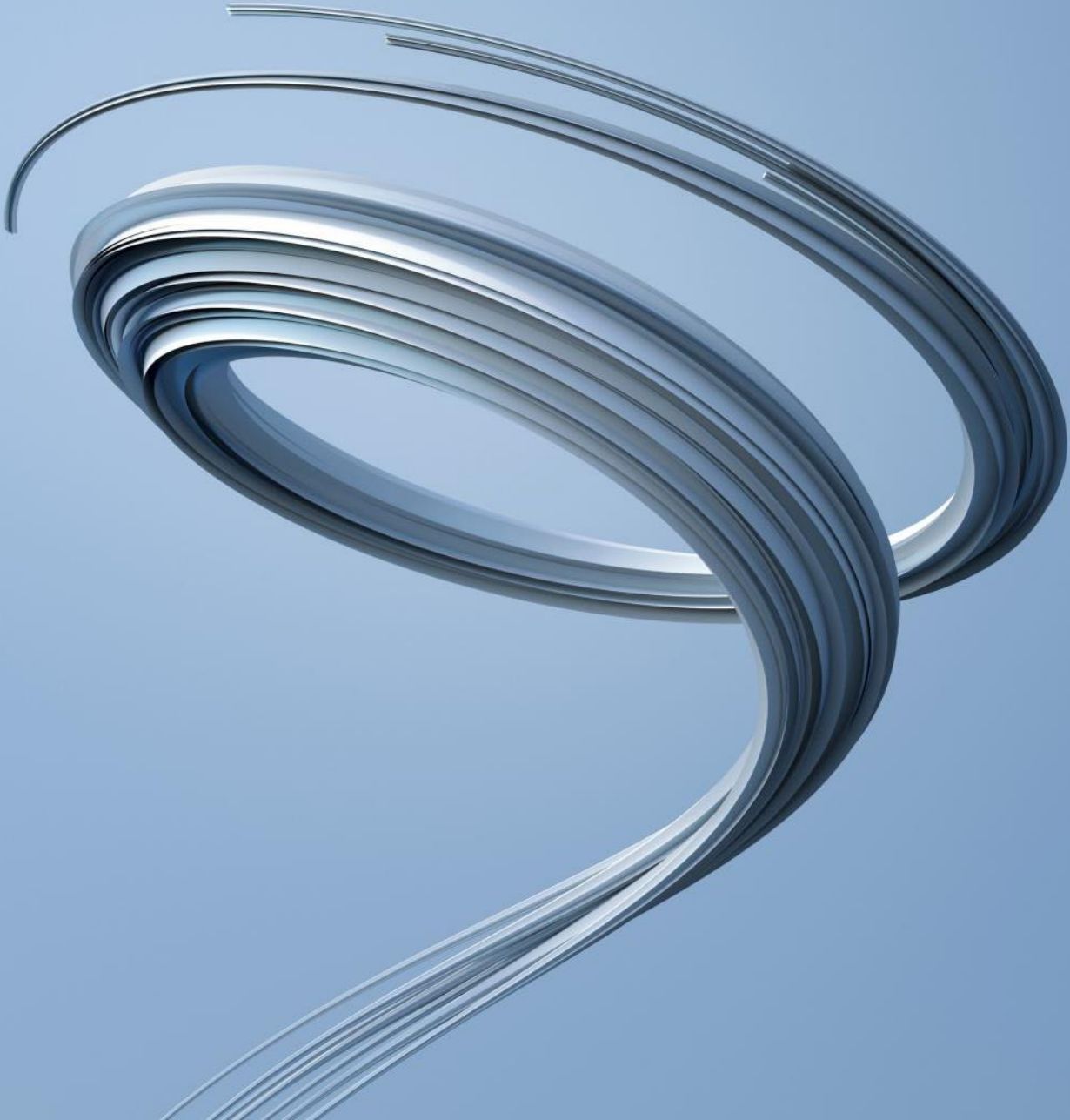


Data breaches



Performance Issues

- Overprovisioning
- VM sprawl (too many unused VMs)
- Resource contention



Cost Management Challenges

- Pay-per-use can spiral
- Unused VMs add cost
- Lack of visibility = billing shocks

Disadvantages of Virtual Machines

- Slower than containers
- Resource overhead
- Complex network setups
- Longer boot times



How to Overcome the Challenges



Use automation tools
(Ansible, Terraform)



Monitor usage and set
alerts



Train staff
continuously



Use hybrid cloud
wisely

Cloud Management Best Practices

- Document configurations
- Use tagging and naming conventions
- Regular cost reviews
- Disaster recovery planning



Summary

- Cloud infrastructure = virtualized compute, storage, and network
- VIMs help manage resources
- VMs offer isolation and flexibility
- But also come with costs and complexity



Final Thoughts

“Managing cloud is like playing chess blindfolded... unless you have the right tools!”

Explore hands-on tools like:

- VirtualBox
- Proxmox
- OpenStack DevStack
- AWS Free Tier

Topic 6 - Migrating into the Cloud

Concepts, Challenges, Integration
& Real-world Considerations





Cloud Migration

- Cloud migration is the process of moving digital business operations into the cloud.
- Involves moving data, applications, and IT processes from on-premises or legacy infrastructure to a cloud-based environment.
- Can be done partially (hybrid) or fully (public/private cloud).

Reasons to Migrate to the Cloud



Scalability: Easily increase or decrease resources.



Cost Efficiency: Pay-as-you-go pricing reduces capital expenditure.



Performance: Improved speed, performance, and uptime.



Disaster Recovery: Built-in redundancy and failover options.



Remote Access: Access from anywhere using internet.

Types of Cloud Services



IaaS (Infrastructure as a Service): Provides virtualized computing resources via the internet (e.g., AWS EC2, Azure VM).



PaaS (Platform as a Service): Offers hardware and software tools for application development (e.g., Google App Engine).



SaaS (Software as a Service): Delivers software applications over the internet (e.g., Google Workspace, Dropbox).

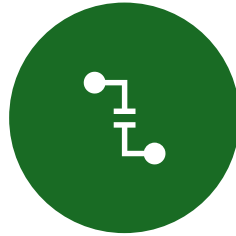
Popular Cloud Providers

- **Amazon Web Services (AWS):** Leader in global cloud market.
- **Microsoft Azure:** Strong in hybrid cloud integration.
- **Google Cloud Platform (GCP):** Known for data analytics and AI tools.
- **IBM Cloud, Oracle Cloud:** Used in enterprise and legacy migration contexts.

Key Challenges in Cloud Migration



DOWNTIME: RISK OF APPLICATION UNAVAILABILITY DURING MIGRATION.



DATA LOSS: IMPROPER HANDLING CAN LEAD TO PERMANENT DATA LOSS.



SECURITY: DATA IN TRANSIT AND AT REST MUST BE PROTECTED.



COMPLIANCE: MUST MEET REGULATORY REQUIREMENTS LIKE GDPR, HIPAA.



COMPATIBILITY: LEGACY APPS MAY NOT RUN SMOOTHLY IN CLOUD.

Planning



Business Goals: Define clear objectives (e.g., cost reduction, scalability).



Assessment: Evaluate existing infrastructure and dependencies.



Cost-Benefit Analysis: Weigh potential savings against migration costs.



Choose Cloud Model: Public, private, or hybrid?

The 6 R's of Cloud Migration



Rehost

Lift and shift.



Refactor

Minor code changes to fit cloud environment.



Revise

Major changes to existing code.



Rebuild

Redesign from scratch.



Replace

Substitute with SaaS solutions.



Retire

Eliminate outdated components.

Rehost (Lift and Shift)

- Move applications without modifying code.
- Typically faster and less expensive.
- Suitable for legacy systems that are hard to refactor.
- **Example:** Moving a virtual machine from a local data center to AWS EC2.

Refactor (Replatform)

- Make minimal changes to adapt the app to the cloud.
- Improve performance or scalability.
- Reuse existing code and frameworks.

Example: Moving an app to a managed database or using container services.

Revise (Re-architect)

- Modify or extend existing code base.
- Required when existing code is not optimized for cloud.
- Prepares for better long-term cloud-native benefits.

Example: Splitting a monolithic app into microservices.

Rebuild (Redesign from Scratch)

- Complete rewrite of the application.
- Highest cost and time investment.
- Best option for outdated or inefficient legacy systems.

Example: Rebuilding an HR system using cloud-native frameworks.

Replace (Drop and Shop)

- Replace your in-house app with SaaS.
- Avoids development cost and effort.
- Fast deployment and maintenance.

Example: Switching to Salesforce instead of running your own CRM.

Retire (Decommissioning)

- Identify and eliminate applications no longer needed.
- Frees up resources.
- Reduces complexity and cost.

Example: Retiring an old reporting tool replaced by BI dashboards in cloud.

Steps to Cloud Migration



**CREATE A
MIGRATION
STRATEGY**



**EVALUATE
APPLICATIONS
AND DATA**



**CHOOSE THE
RIGHT CLOUD
PROVIDER**



**PREPARE THE
ENVIRONMENT**



MIGRATE AND TEST



**MONITOR AND
OPTIMIZE**

Step 1 - Create a Migration Strategy



Define the business objectives for moving to the cloud.



Determine scope, timeline, and budget.



Identify stakeholders and assign responsibilities.

Step 2 - Evaluate Applications and Data



Audit existing infrastructure.



Classify applications based on criticality and complexity.



Identify dependencies and compliance requirements.

Step 3 - Choose the Right Cloud Provider



Compare pricing models, SLAs, and services offered.



Consider support, security features, and compliance certifications.



Evaluate existing provider integrations and ecosystem compatibility.

Step 4 - Prepare the Environment

Set up

- Set up accounts, access policies, and permissions.

Design

- Design network architecture and storage.

Implement

- Implement identity and access management (IAM).

Step 5 - Migrate and Test



Begin with low-risk applications as a pilot.



Execute the migration using tools and automation.



Validate the system functionality through rigorous testing.

Step 6 - Monitor and Optimize



Monitor usage, performance, and security.



Tune systems for cost and performance efficiency.



Implement alerts and analytics dashboards.

Migration Tools & Platforms

- **AWS Migration Hub**
- **Azure Migrate**
- **Google Migrate for Compute Engine**
- **CloudEndure, Velostrata**

Limitations That Deter Adoption



Poor Connectivity: Limits real-time access.



Legacy Dependencies: Older apps may resist change.



Lack of Cloud Talent: Skilled professionals are in demand.



Concerns over Vendor Lock-in

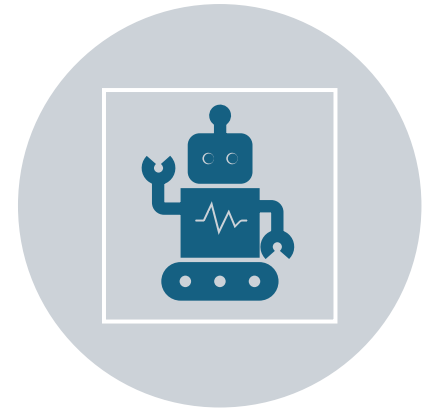
Cloud Integration Types



DATA INTEGRATION: SYNCHRONIZE
DATA ACROSS SYSTEMS.



APPLICATION INTEGRATION:
CONNECT DISPARATE APPS FOR
WORKFLOW.



PROCESS INTEGRATION: END-TO-
END BUSINESS PROCESS
AUTOMATION.

Data Integration

- Combines data from different sources into a single, unified view.
- Supports real-time or batch synchronization between databases and storage.
- Ensures consistency, quality, and availability of business-critical data.

Example: Syncing product inventory data between a cloud-based e-commerce platform and an on-premises ERP system.

Application Integration

- Connects different applications to work together.
- Facilitates communication and data exchange across platforms.
- Enables automation of workflows that span multiple systems.

Example: Integrating Salesforce (CRM) with Mailchimp (Email Marketing) to auto-send customer emails after a sales update.

Process Integration

- Aligns business processes across applications and departments.
- Streamlines multi-step workflows that depend on input/output from different systems.
- Often managed through orchestration or Business Process Management (BPM) tools.

Example: An order placed on an e-commerce platform triggers inventory check, invoicing, and shipment tracking—spanning several apps.

Requirements for Integration



Secure APIs



Real-time and Batch Processing Support



Data Transformation



Monitoring and Logging

Attributes of Integration Platforms

Scalability:

- Must handle growing data.

Interoperability:

- Work with multiple environments.

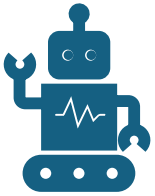
Security:

- Data encryption and compliance support.

Ease of Use:

- Low-code/drag-drop preferred.

What is Jitterbit Integration Server?



**Low-code iPaaS
(Integration Platform as a
Service)**



Supports connecting cloud
and on-premise systems.



Visual data mapping and
transformation.



Real-time and scheduled
integrations.



Risks in Cloud Migration

- **Vendor Lock-in**
- **Budget Overruns**
- **Service Disruptions**
- **Loss of Governance**
- **Incompatibility**



Mitigation Strategies

- **Use Open Standards/APIs**
- **Document Everything**
- **Start with Pilot Projects**
- **Regular Testing and Backup**

Post-Migration Considerations



**MONITORING
AND LOGGING**



**PERFORMANCE
OPTIMIZATION**



**SECURITY POLICY
UPDATES**

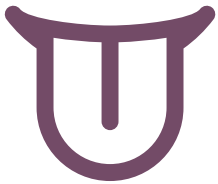


USER TRAINING

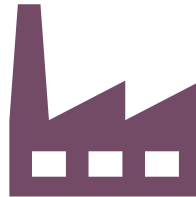


PERIODIC AUDITS

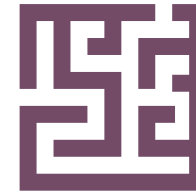
Cloud Messaging Queuing



Message Queuing:
Temporarily holds messages
for processing.



Decouples Components:
Producer and consumer run
independently.



Improves Reliability:
Handles spikes without
overloading systems.

Examples of Messaging Queuing Tools



Amazon SQS: Reliable queue service for distributed apps.



Apache Kafka: High-throughput distributed messaging.



RabbitMQ: Lightweight and flexible broker.



Azure Queue Storage

Summary of Key Points



CLOUD MIGRATION
IS COMPLEX BUT
REWARDING



CHOOSE THE RIGHT
MODEL AND TOOLS



INTEGRATE
THOUGHTFULLY



BE AWARE OF RISKS
AND MITIGATE THEM



USE MESSAGING
QUEUES FOR BETTER
ARCHITECTURE

Topic 7 - Infrastructure as a Service (IaaS) Providers

Exploring Features, Technologies, and Leading Providers

Introduction to IaaS

- Infrastructure as a Service (IaaS) is a form of cloud computing that provides virtualized computing resources over the internet. With IaaS, businesses can rent servers, storage, and networking infrastructure on-demand, rather than owning and maintaining physical data centers. It provides maximum control over operating systems, storage, and deployed applications, making it ideal for system administrators and developers.
- This model lays the foundation for understanding how modern cloud infrastructures are structured and consumed, leading into a comparison with other cloud service models.

IaaS in Cloud Models

- IaaS is one of the three main cloud service models: IaaS, PaaS, and SaaS. It offers the highest level of flexibility and management control.
- Understanding IaaS's position helps us appreciate the unique features that make it suitable for specific use cases, which we will now explore.

Key Features of IaaS



On-demand Self-Service: Users can provision resources as needed.



Resource Pooling: Providers pool computing resources to serve multiple clients.



Rapid Elasticity: Scale resources up or down quickly.



Measured Service: Pay only for what you use. These features enable agility and cost-efficiency.

Elasticity and Scalability

- Elasticity refers to the ability of a system to automatically scale in or out depending on demand. For example, during a major sales event, an e-commerce site can automatically add servers to handle increased traffic, and reduce them once traffic returns to normal—ensuring cost optimization and performance stability.
- To enable this kind of flexibility, IaaS depends heavily on virtualization.

Virtualization Technology

- IaaS heavily relies on virtualization to abstract hardware resources. Virtual machines (VMs) are created on physical servers using hypervisors such as VMware ESXi, Xen, KVM, or Hyper-V. This enables multiple users to share the same physical hardware securely and efficiently, reducing hardware costs and improving scalability.
- Since IaaS infrastructure spans the globe, let's now look at the importance of where this infrastructure is physically located.

Geographic Presence

- Most IaaS providers have a global network of data centers. These are grouped into geographic regions and availability zones. The wider the presence, the more options a business has to place data closer to users—reducing latency and meeting data residency laws.
- But having global data centers alone isn't enough—strategic placement plays a crucial role, as we'll now see.

Importance of Data Center Location

- Choosing a data center location is crucial for performance and compliance. For instance, a company targeting Asian customers might choose a Singapore or Tokyo region. Data locality also affects legal compliance, e.g., GDPR compliance in Europe.
- Once a location is selected, users need methods to interact with and manage their infrastructure.

User Interfaces and Access to Servers

- Web-based dashboards (e.g., AWS Management Console)
- Command-line tools (e.g., Azure CLI, AWS CLI)
- RESTful APIs and SDKs for automation This variety ensures accessibility for both beginners and advanced users.

Interfaces in Practice

- For example, DevOps teams often use APIs or tools like Terraform to automatically create and manage infrastructure as code. This automation improves consistency and reduces human error in large deployments.
- Let's now consider how some businesses plan ahead and save costs through capacity reservation.

Advanced Reservation of Capacity

- IaaS providers often offer reserved instances at a lower cost compared to on-demand instances. This is useful when you have predictable workloads. For instance, a business with a steady monthly traffic pattern can reserve capacity in advance and save up to 70%.
- To see how this works in practice, consider a real-world academic use case.

Real-World Example of Reservation

- Imagine a university hosting an exam portal every semester. Since they know the exact dates of high usage, they can reserve capacity in advance, ensuring both cost savings and availability.
- But what happens when usage patterns are unpredictable? That's where auto scaling and load balancing come in.

Auto Scaling and Load Balancing

- Auto Scaling automatically adjusts the number of VM instances to meet demand. Load Balancing distributes traffic evenly across instances. Together, they ensure high availability, fault tolerance, and cost efficiency during variable load conditions.
- This is especially crucial for applications facing sudden traffic spikes, like e-commerce websites.

E-commerce Example

- During flash sales, a fashion website may receive traffic spikes. Auto scaling adds more servers to handle the surge, and the load balancer routes users across these servers to prevent crashes, ensuring a smooth customer experience.
- To guarantee performance, these services are backed by formal agreements known as SLAs.

Understanding SLAs

- Service-Level Agreements define the expected level of service, typically including uptime (e.g., 99.99%), response times, and support levels. SLAs build trust and set accountability between provider and customer. Breaching SLAs may result in service credits.
- The implications of these agreements are significant—especially for mission-critical applications.

SLA Implications

- If a provider promises 99.99% uptime, this means only about 52 minutes of downtime per year is acceptable. Applications requiring high availability, like online banking, must evaluate SLAs critically before choosing a provider.
- Underpinning these performance commitments is the technology that manages VMs—hypervisors.

Hypervisors Overview

Hypervisors manage VMs and are classified into:

- Type 1: Bare-metal (e.g., Xen, VMware ESXi)
- Type 2: Hosted (e.g., VirtualBox) Type 1 hypervisors are more secure and commonly used in data centers. AWS uses a customized version of Xen and Nitro.

Hypervisors enable the use of multiple operating systems.

Operating System Choices

- Users can install any OS, such as Ubuntu, CentOS, Red Hat, or Windows Server. This flexibility supports a wide variety of use cases—from hosting websites on Linux to running legacy enterprise apps on Windows.
- Now that we understand the core technology, let's examine the major players offering these services.

Introduction to IaaS Providers

Several key players dominate the IaaS space, each with unique strengths:

- Amazon Web Services (AWS)
- Flexiscale
- Joyent
- Rackspace We'll now explore each in more detail to compare what they bring to the table.

Amazon Web Services (AWS)

- AWS is the largest and most widely used IaaS provider. It offers a vast array of services beyond compute and storage, including AI, IoT, and big data analytics. AWS supports global infrastructure with 30+ geographic regions.
- Let's take a closer look at one of AWS's flagship IaaS offerings—Amazon EC2.



Amazon EC2 Features

Elastic Compute Cloud (EC2) allows you to launch virtual servers on-demand. Features include:

- Variety of instance types (General, Compute, GPU)
- Elastic IPs
- AMIs (Amazon Machine Images)
- Integration with Auto Scaling, ELB, CloudWatch



Amazon
EC2

EC2 Use Cases

- A startup can launch a web app with minimal upfront cost.
- Research teams can run machine learning models on spot instances.
- Enterprises can deploy scalable microservices architecture.

Beyond AWS, there are other notable providers with unique features worth examining.

Flexiscale Overview

Flexiscale was among the first cloud IaaS platforms. It emphasized ease of use, GUI-based management, and real-time provisioning. It laid the groundwork for future European cloud innovations.



Joyent Overview

Joyent provides SmartOS-based virtual machines optimized for real-time apps. Known for supporting Node.js early, Joyent is now part of Samsung and focuses on cloud-native workloads and high-performance computing.



Rackspace Cloud Servers

- Rackspace specializes in managed cloud services. It helps businesses deploy and manage cloud infrastructure on platforms like AWS, Azure, and Google Cloud. Their focus is on customer support, making them ideal for companies with limited IT staff.



Comparing Providers

Feature	AWS	Flexiscale	Joyent	Rackspace
Global Reach	High	Low	Moderate	High
Support	Good	Moderate	Developer-focused	High-touch
Pricing Flexibility	High	Medium	Medium	Low
OS Flexibility	Broad	Broad	SmartOS	Broad

Choosing the Right IaaS

- Cost predictability
- Performance and availability needs
- Compliance requirements
- In-house expertise A healthcare startup might choose AWS for scalability, while a small firm might pick Rackspace for its support.

Real-World Applications of IaaS

- Education: Virtual labs, student VMs
 - E-commerce: Scalable web stores
 - Media: Video rendering farms
 - Government: Secure data processing
- These use cases highlight the flexibility of IaaS across sectors.

Challenges with IaaS

- Cost overruns without monitoring
- Data security risks if misconfigured
- Vendor lock-in
- Complexity in setup and scaling These should be planned for during adoption.